Foreign Currency Borrowing and the Investment Channel of Monetary Policy [*]

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Abstract

This paper studies how foreign currency debt affects firms' investment response to monetary policy. Using firm-level data on foreign currency bond issuance across 15 countries, I document that highly leveraged firms with high foreign currency debt respond less to monetary easing than firms with similar leverage and less foreign currency debt. For a firm at the 75th percentiles of both leverage and foreign currency debt share, investment falls by 3.5 percentage points eight quarters after a one percentage point easing, while it rises for a highly leveraged firm with little foreign currency debt. A heterogeneous-firm small open economy model in which firms choose the currency of their debt reproduces this pattern. I find that while dollar debt dampens the aggregate investment response to easing only mildly, leaning against depreciations is costly: after a foreign monetary tightening, the more the central bank defends the currency, the larger the fall in investment.

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1 Introduction

Monetary easing can have a weaker effect on investment when highly leveraged firms owe substantial debt in foreign currency. Lower interest rates reduce borrowing costs, stimulate demand, and depreciate the exchange rate, boosting competitiveness. Yet for firms with substantial foreign currency debt, a depreciation raises the domestic value of their liabilities, potentially offsetting or in extreme cases, even reversing the expansionary effects. This paper asks: do firms with higher foreign currency exposure and leverage respond less to a monetary easing? The question is not only of academic interest but also of pressing policy relevance. Episodes such as Turkey's 2018–2019 "balance sheet recession," in which rapid currency depreciation inflated foreign liabilities and triggered a contraction in credit and investment, highlight how currency mismatches can undermine the effectiveness of monetary easing.

Understanding this interaction is critical given the prevalence of foreign currency borrowing in emerging markets and the global dominance of the U.S. dollar in trade and finance. Figure 1 illustrates the extent of foreign currency borrowing across countries and the central role of the dollar in corporate debt issuance. Total debt and total foreign debt outstanding in the corporate sector has increased steadily from 2000 to 2020. While the share of foreign currency debt to total debt has dropped, it still remains high at around 40% in recent years. Table 1 also shows that foreign borrowing is also predominantly denominated in dollars and a few other global currencies, suggesting that exchange rate movements in a few currencies has implications for the majority of corporate sector balance sheets [1]. Thus, monetary easing may be less effective—and even contractionary—for heavily indebted firms with foreign currency exposure.

The existing literature has documented both the potential benefits and costs of foreign currency borrowing. On one hand, firms borrow in foreign currency to take advantage of lower interest rates and better market access. On the other, this creates balance sheet

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[1] Figure 1 and Table 1 is showing data for all non-US corporate firms (except financial, public and utilities firms) who issue bonds in the the SDC Platinum New Issues dataset. This includes any bond issuance with a maturity of at least one year by any publically listed firm. The data covers publically listed firms from all non-US countries.

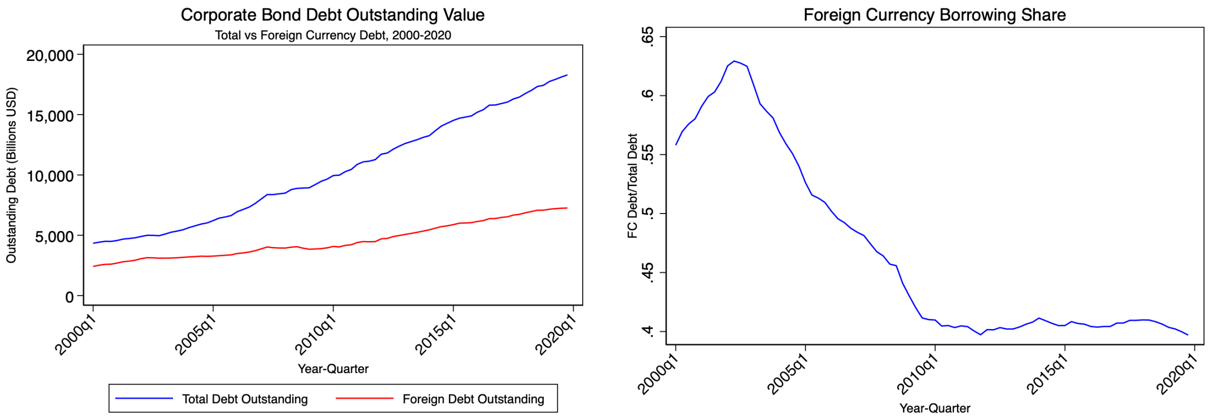


Figure 1. Total and foreign debt outstanding by corporate sector. Debt outstanding is calculated from SDC Platinum Global New Issues using all corporate bonds issued from the year 2000 to 2020. I include corporate bonds issued across *all* non-US corporate firms wheras my analysis uses a sample, detailed further in Section 3.

risk when exchange rates move. Most prior work has studied the consequences of exchange rate depreciations as isolated shocks—often linked to external crises or global risk episodes—rather than as part of domestic monetary policy actions. This distinction matters: monetary policy shocks affect the economy not only through exchange rates but also through domestic demand, credit conditions, and expectations. As a result, the net impact on firms with currency mismatches is ambiguous ex ante. Whether the demand-side stimulus outweighs the balance sheet hit is an empirical question, and one that has not been systematically examined at the firm level across countries.

Table 1. Foreign Bond Issuance by Currency Denomination

Currency	**Percentage**	**Cumulative Percentage**
US Dollar	51.44%	51.44%
Euro	8.99%	60.43%
Hong Kong Dollar	6.52%	66.95%
Japanese Yen	6.36%	73.31%
Swiss Franc	3.83%	77.14%
Other	22.86%	100.00%
Total	100.00%	100.00%

I address this question using a new panel dataset that matches firm-level financial statements from Worldscope with detailed corporate bond issuance records from SDC Platinum for 15 advanced and emerging economies between 2000 and 2019. This allows me to construct a quarterly measure of each firm's foreign currency debt share based on outstanding bonds, complemented by standard balance sheet variables such as leverage, liquidity, and size. I combine this micro data with identified monetary policy shocks from Choi et al. (2024), aggregated to the quarterly level, which isolate exogenous changes in domestic monetary policy.

My empirical strategy follows a local projection framework that interacts monetary policy shocks with firm-level foreign currency debt shares and leverage. This design allows me to estimate the differential investment responses of firms with varying degrees of currency mismatch and financial vulnerability, while controlling for firm characteristics, sectoral trends, and macroeconomic conditions.

I find that firms with both high leverage and high foreign currency debt respond significantly less to a monetary easing than otherwise similar firms. In some cases, the effect is not just weaker—it is negative: a one percentage point monetary easing reduces investment in these firms over the subsequent year. By contrast, highly leveraged firms with predominantly domestic currency debt increase investment following an easing, consistent with the traditional credit channel. The adverse effect of easing for high foreign currency–exposed, highly leveraged firms grows over time, peaking around 4–8 quarters after the policy shock.

The heterogeneous responses to monetary policy documented in this paper imply that the aggregate impact of policy in economies with currency mismatches depends on the distribution of firms across leverage and foreign currency exposure. In my data, larger and faster-growing firms are more likely to select into foreign currency borrowing, often taking on substantial shares of their total debt in foreign currency. When these firms are also highly leveraged, the depreciation associated with monetary easing can generate adverse balance sheet effects that offset or even outweigh the expansionary forces from lower interest rates and stronger demand.

To study the implications of these results for aggregate outcomes and for policy, I develop a small open economy New Keynesian model with heterogeneous firms that choose how much to invest, how much to borrow, and in which currency to borrow.

Dollar debt is cheaper at the margin but requires a fixed cost to access, so larger firms select into it, as in the data. A monetary easing lowers domestic interest rates and depreciates the peso, which raises the real value of firms' dollar debt and lowers their internal funds. Because equity is costly, the loss reduces investment for firms whose marginal value of internal funds exceeds one, and the reduction rises with leverage and the dollar share of debt. Calibrated to the cross-section of firms in my sample, the model reproduces the signs of the empirical estimates. I then use the model for two counterfactuals. First, removing dollar debt raises the investment response to an easing by about an eighth on impact, while the peso depreciates by less, because dollar repayments no longer have to be financed through the trade balance. Second, after a foreign monetary policy tightening, the more the central bank leans against the resulting depreciation, the larger the domestic contraction. Under a nominal exchange rate peg, investment falls by 0.41 percent on impact, compared with a small decline under an inflation-targeting rule, because the balance-sheet cost of the depreciation is small relative to its boost to exports.

2 Related Literature

The contribution of this paper is to show how the currency composition of corporate debt shapes firms' investment responses to domestic monetary policy. Three strands of research inform this question: evidence on the currency composition of external portfolios, studies of firms' foreign currency borrowing decisions, and work on monetary policy transmission. The first documents a decline in aggregate foreign currency liability exposure, even as vulnerabilities persist in portfolio debt. My data show a related pattern: the level of foreign currency corporate bond debt has risen over time, while its share of outstanding corporate bond debt has fallen. This pattern aligns with Allen (2023), who document that many economies, especially emerging markets, have shifted from short to long foreign currency positions as the weight of foreign currency liabilities on external balance sheets declined. While their paper focuses on foreign currency exposure of countries, my paper focuses more on the foreign currency exposure of private sector corporations. Bruno and Shin (2017) document that firms with high cash holdings are more likely to issue US dollar-denominated bonds, particularly during periods when the dollar carry trade is more favorable, suggesting that firms engage in financial speculation rather than purely operational financing. This carry trade behavior has important implications for monetary policy, as firms essentially act as financial intermediaries, borrowing in foreign currency to exploit interest rate differentials. Maggiori et al. (2020) establish that investor holdings are biased toward their own currencies to such an extent that countries typically hold most of the foreign-debt securities denominated in their currency, with the notable exception of US dollar bonds where foreign investors hold significant shares. Consequently, non-US firms largely issue U.S dollar denominated debt when borrowing abroad. Gozzi et al. (2010) show that while a large fraction of capital raisings occurs in international markets, a very small number of firms accounts for the bulk of international capital raisings. These patterns reveal that access to international capital markets is highly concentrated among large firms, creating heterogeneous exposure to global financial conditions across the firm size distribution. Similarly, my paper begins by documenting that firms that issue foreign currency debt are larger and high growth firms.

My paper is also related to the literature that studies the endogenous choice of foreign currency external financing by the non-financial corporate sector and the consequences of this currency mismatch. Salomao and Varela (2021) shows that foreign currency borrowing emerges from a trade-off between cheaper funding and default risk that emerges from exchange rate fluctuations, with more productive firms selecting into foreign currency debt. Alternatively, Eren et al. (2022) demonstrate that firms may use currency mismatches to signal quality under information asymmetries, while Gopinath and Stein (2020) explain how the dollar's dominance in trade and finance leads emerging market firms to endogenously take on currency mismatches. The selection pattern in my data is close to that in Maggiori et al. (2023), where larger and more productive firms select into foreign currency borrowing to reduce overall borrowing costs. In their model, issuing debt in a foreign currency can be provide firms with lower interest rates, but entail large sunk costs. Empirically, Kalemli-Ozcan et al. (2021) document that exchange rate movements affect firm leverage through balance sheet effects - appreciations allow firms with foreign currency debt to borrow more, while depreciations force deleveraging, with asymmetrically larger effects during depreciations. I contribute to this literature by examining whether currency mismatches alter the effects of domestic monetary policy. While prior work has primarily analyzed the consequences of exchange rate depreciations in isolation, my focus is on monetary policy shocks, which can affect exchange rates (see Eichenbaum and Evans (1995), Inoue and Rossi (2019)) and stimulate the economy through improved collateral values, expenditure switching effects, and decreased debt burdens on local currency debt.

The literature on firm heterogeneity and monetary policy transmission has established that firm characteristics shape how monetary policy affects investment decisions. Gertler and Gilchrist (1994) provide evidence that small firms account for a disproportionate share of the manufacturing decline following monetary tightening, highlighting the role of financial frictions in propagating monetary shocks. More recent work has challenged and refined this view. Ottonello and Winberry (2020) find that firms with low default risk—those with low debt burdens and high distance to default—are actually the most responsive to monetary shocks, as they face a flatter marginal cost curve for financing investment. My paper extends this literature by identifying an additional channel through which firm heterogeneity affects monetary policy transmission in open economy settings. I demonstrate that in open economies, the currency composition of debt creates an additional layer of heterogeneity. Highly leveraged firms with substantial foreign currency debt increase investment significantly less following monetary easing compared to similarly leveraged firms with only domestic currency debt.

Finally, my paper relates to the literature that studies monetary policy transmission in open economies. Most closely, Céspedes et al. (2004) show that, even in the presence of currency mismatches that raise the domestic value of liabilities after a depreciation, monetary policy remains expansionary because the expenditure-switching effect dominates the balance-sheet channel. Gertler et al. (2007) embed financial frictions into an open-economy New Keynesian framework, showing how external constraints amplify the effects of monetary policy, including an extension where entrepreneurs borrow in foreign currency—although without an endogenous choice into such borrowing. More recently, Auclert et al. (2021b) introduce household heterogeneity into a small open economy model, emphasizing a "real income channel" whereby depreciations reduce real incomes through higher import prices, potentially making monetary policy less expansionary when trade elasticities are low. I contribute to this literature by empirically and theoretically studying firm-level heterogeneity in response to monetary policy, focusing on how investment reacts when firms endogenously choose foreign currency debt and thus differ in their exposure to the balance-sheet and expenditure-switching effects of a policy-induced depreciation.

The rest of the paper is organized as follows. Section 3 describes the sources of data and details on the data construction. Section 4 details my empiricial approach and results. Section 5 concludes.

3 Data Description

Foreign currency borrowing in my sample is concentrated among larger firms and varies substantially across countries and sectors. This section describes how I measure that exposure, construct the firm-level panel, and match it to monetary policy shocks. It then documents the sample patterns that motivate the empirical analysis.

I use quarterly Worldscope Fundamentals data provided by LSEG. Worldscope contains standardized balance sheet, income statement, and cash flow data for firms in over 120 advanced and emerging economies, with coverage extending back to the 1980s for many firms. The database focuses primarily on large publicly traded companies. My baseline sample construction follows standard practices in the international corporate finance literature. I exclude firms in the financial, utilities, and the public sector. Ideally, I would focus on a firms in a broader set of emerging market economies but Worldscope's coverage and match quality with SDC Platinum leads me to consider firms in 15 countries in both advanced and emerging market economies.

I require firms to have at least 24 quarterly observations to ensure sufficient time-series variation for the local projection analysis. I also impose standard data quality filters, dropping observations with negative total assets, sales, leverage ratios, or liquidity measures.

I construct firm-level capital stock by setting the initial capital stock equal to gross property, plant, and equipment when first available, and then use the accumulation equation $K_{i,t+1} = K_{it} + (\text{Net PPE}_{it} - \text{Net PPE}_{i,t-1})$ for subsequent periods. Investment rates are calculated as log changes in the capital stock: $\text{inv}_{i,t+h} = \log(K_{i,t+h}/K_{i,t})$.

I define leverage as total debt scaled by total assets. I winsorize leverage and sales growth at the 0.5th and 99.5th percentiles to mitigate the influence of outliers. All balance sheet variables are converted to U.S. dollars using end-of-quarter exchange rates from Datastream, then deflated to real terms using country-specific GDP deflators from the IMF World Economic Outlook database.

While Worldscope provides excellent coverage of large public companies, it has several limitations. First, Worldscope primarily covers publicly traded firms, potentially missing private companies that also access international bond markets. Coverage skews toward larger firms, which may have different financing patterns than smaller companies. Second, some emerging market economies have limited historical coverage or a limited number of firms in the dataset and are therefore not in my sample. Finally, while Worldscope does have a variable that identifies exporter status, most entries are missing. While I recognize that export status is an important predictor of selection into and the balance sheet effects of foreign currency borrowing, using the export status variable reduces the sample size drastically. I include a version of my baseline regression in the appendix with firm fixed effects as a robustness check.

3.1 Corporate Bond Issuance Data

SDC Platinum Global New Issues provides transaction-level data on corporate bond issuance worldwide. I use these data to construct a firm-level proxy for foreign exchange exposure. While comprehensive balance sheet data on all foreign currency liabilities would be ideal, such granular firm level information is not systematically available across countries and time. I therefore use SDC Platinum's detailed bond issuance records to construct a measure of firms' foreign currency debt exposure.

Corporate bonds represent a significant component of firms' external financing, particularly for the large public companies in my Worldscope sample. While this approach neglects bank loans, trade credit, and other foreign currency liabilities, bond debt provides precise currency denomination information and detailed maturity structures allowing me to track outstanding debt stocks over time.

I aggregate individual bond issuances to create firm-quarter measures of total and foreign currency debt outstanding. My approach accounts for the complete lifecycle of each bond by adding the face value of the bond to the firms total debt stock when the bond is issued, and remains on the firms balance sheet until maturity or early redemptions/calls. I classify debt as a foreign currency issuance if the debt is denominated in a different currency then the local currency of the firm's nation of incorporation. For example, bonds issued by Brazilian firms are classified as foreign currency unless denominated in Brazilian Real. I account for historical currency changes (Brazilian currency reforms, Turkish lira redenomination). Finally, I aggregate the face value of foreign and total bond issuances for each firm at the year-quarter level. Since I do not have a clear measure of the initial total or foreign outstanding debt from SDC Global New Issues, I collect all bond issuances for all corporate firms starting in 1965. I assume the first bond issuance in the historical time series for an individual firm is the initial debt stock for that firm.

Following standard practice, I consolidate all bond issuance activity at the ultimate parent company level using SDC's ultimate parent CUSIP identifiers. This approach captures the complete bond issuance of corporate groups, including activity by both financial and nonfinancial subsidiaries. SDC defines ultimate parent companies using a 50% ownership threshold with hierarchical consolidation. This consolidation ensures my foreign currency exposure measures reflect the true economic exposure of corporate groups rather than being fragmented across legal entities, while maintaining my focus on nonfinancial corporate behavior.

3.2 Monetary Policy Shock Data

I use monetary policy shock data available from Choi et al. (2024). They provide monthy identified monetary policy shock for each country. They collect monetary policy shocks from various sources, using a hierarchical approach where identification is based on high frequency, swap yields, bond yields, survey suprises, and taylor rule residuals in that order. I aggregate the monthly monetary policy shocks to the quarter level for each country. In all of the analysis to follow, I normalize the monetary policy shock so that positive values indicate monetary expansion.

3.3 Summary Statistics

Table 2 presents summary statistics for the 15 countries in my sample, revealing substantial heterogeneity in foreign currency borrowing patterns across both advanced and emerging market economies. The sample comprises 1,019 firms with 29,223 firm-quarter observations from 2000 to 2019, though most countries enter the sample in 2007. The column FX issuer is the share of firms that have ever issued foreign currency bonds throughout the entire sample, while FX share is the share of total outstanding debt issued in foreign currency relative to all total outstanding debt amongst firms that have ever issued foreign bonds. The most striking feature of the data is the dramatic cross-country variation in foreign currency debt exposure. The mean share of interest-bearing debt denominated in foreign currency ranges from 4.8% for South Korean firms to 97.8% for Chilean firms. Advanced economies display similarly diverse patterns: while Japanese firms hold 36.5% of their debt in foreign currency on average, UK firms maintain 89.0% foreign currency debt shares, approaching levels seen in many emerging markets. The extensive margin of foreign currency borrowing—captured by the share of firms who have ever issued foreign currency debt displays comparable heterogeneity. In Korea, only 3.2% of firms issue foreign currency bonds, while in Chile this figure reaches 78.6%. Notably, the relationship between the intensive and extensive margins varies across countries. For instance, Brazilian firms show high average foreign currency debt shares (82.3%) but moderate participation rates (37.5%), suggesting that firms who do

Table 2. Summary statistics of country-level variables

Market	Nation	Obs.	Firms	FX share*	FX issuer†	Min yr	Max yr	MPS Source
EME	Brazil	1825	64	0.823	0.375	2007	2019	High Frequency
	Chile	1053	28	0.978	0.786	2007	2019	Taylor Residual
	India	253	26	0.846	0.192	2009	2019	High Frequency
	Indonesia	769	28	0.560	0.357	2007	2019	Taylor Residual
	Malaysia	1852	68	0.214	0.147	2007	2019	Taylor Residual
	Mexico	1331	53	0.765	0.472	2000	2019	Swap Yields, Taylor Residual
	Philippines	418	15	0.399	0.267	2007	2019	Swap Yields, Taylor Residual
	Russian Federation	719	24	0.725	0.417	2007	2019	Taylor Residual
	Thailand	1010	48	0.184	0.063	2007	2019	Taylor Residual
Developed	Canada	3985	128	0.694	0.523	2000	2019	High Frequency
	France	605	23	0.475	0.522	2007	2019	High Frequency
	Germany	2095	69	0.382	0.464	2007	2019	High Frequency, Bond Yields
	Japan	11073	317	0.365	0.114	2007	2019	High Frequency
	Korea (South)	1507	95	0.048	0.032	2007	2019	High Frequency
	United Kingdom	728	33	0.890	0.727	2007	2019	High Frequency, Swap Yield
	All countries	29,223	1,019	0.537	0.282	—	—	

Notes: FX share* is the mean fraction of total interest-bearing debt denominated in foreign currency. FX issuer† is the share of firm–quarters in which the firm has any FX-denominated bond or loan outstanding.

borrow in foreign currency tend to do so heavily. In contrast, Canadian firms display both high foreign currency debt shares (69.4%) and participation rates (52.3%), indicating widespread and intensive use of foreign currency financing.

Table 3 reveals substantial sectoral heterogeneity in foreign currency borrowing that closely aligns with tradability. Mining firms stand out with the highest foreign currency participation and debt shares, followed by agriculture—both sectors with significant export exposure. In contrast, non-tradable sectors like construction, retail trade, and services exhibit markedly lower foreign currency borrowing, despite maintaining comparable leverage ratios. Manufacturing, comprising nearly half my sample, occupies an intermediate position, likely reflecting within-sector heterogeneity between export-oriented and domestic-focused producers.

This sectoral variation provides important additional identification for my empirical analysis. The concentration of foreign currency borrowing among tradable sectors suggests that firms' currency choices reflect operational considerations beyond pure financial arbitrage. Moreover, the stark differences across sectors—with foreign currency debt shares varying by a factor of six between construction and mining—imply that monetary policy transmission through balance sheet effects will exhibit significant sectoral heterogeneity. This pattern motivates my inclusion of sector-time fixed effects in the baseline specification and suggests that the aggregate effects of monetary policy may depend on the sectoral composition of foreign currency borrowing within each economy.

Table 3. Summary Statistics by Sector

Sector	Number of Firms	FX Issuer Share	Leverage	FX Share
Agriculture	9	0.444	0.409	0.403
Mining	78	0.590	0.279	0.679
Construction	100	0.190	0.330	0.114
Manufacturing	484	0.254	0.293	0.238
Transportation	163	0.313	0.412	0.251
Wholesale trade	42	0.381	0.352	0.173
Retail trade	54	0.167	0.297	0.198
Services	89	0.213	0.323	0.212

Table 4 presents firm-level summary statistics, revealing substantial heterogeneity in foreign currency exposure and other financial characteristics within my sample. The foreign currency debt share exhibits particularly stark variation across firms. While the median firm holds no foreign currency debt, firms at the 95th percentile are entirely foreign currency financed. This bimodal pattern suggests that foreign currency borrowing decisions may involve significant fixed costs or threshold effects, leading firms to either abstain entirely or commit substantially to foreign currency financing. Investment rates, measured as log changes in capital stock, average 1.0% quarterly with considerable volatility. This high variance in investment behavior provides the necessary variation to identify differential responses to monetary policy shocks based on firms' currency exposure. The interaction between investment volatility and heterogeneous foreign currency positions forms the core of my identification strategy.

Table 5 reveals differences between firms that access foreign currency debt markets and those that rely exclusively on local currency financing. Foreign currency issuers represent approximately one-third of my sample, but these firms are systematically different along multiple dimensions.

Most notably, foreign currency issuers are substantially larger than their local currency counterparts as measured by real assets. Table 5 shows that the mean and median size of

Table 4. Summary statistics of firm-level variables

Variable	Mean	Median	Std. Dev.	95th Pct.	Obs.
$\log(\Delta k_{i,t+1})$	0.010	0.011	0.169	0.134	29,198
FX Share	0.259	0.000	0.398	1.000	29,198
Leverage	0.321	0.303	0.157	0.596	29,198
log(Real Assets)	8.211	8.272	1.831	11.135	29,198
Real sales growth	0.016	0.019	0.244	0.300	29,198
Liquidity	0.103	0.084	0.085	0.263	29,198

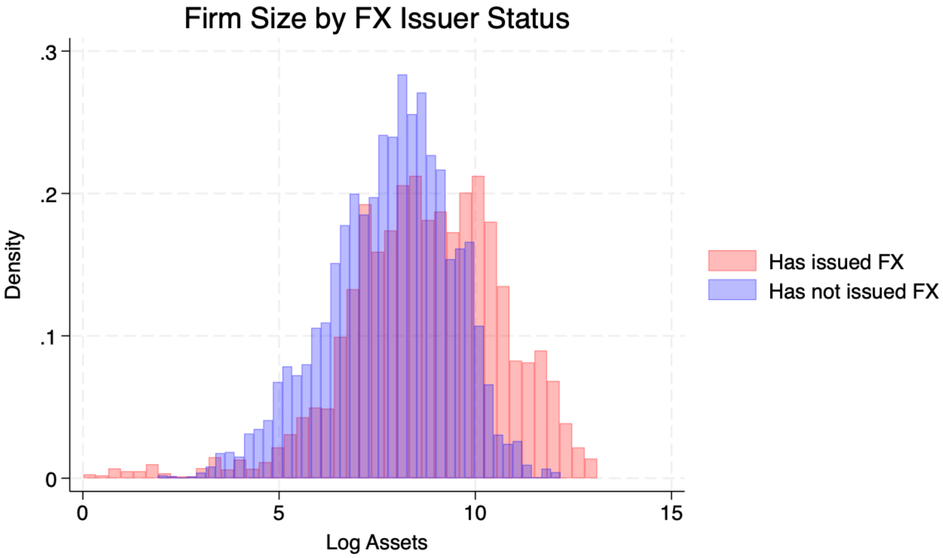


Figure 2. Firm size by FX issuer history. I define Has issued FX as firms that have ever issued debt in a foreign currency at any time in the 2000-2020 sample period.

a firm that issues foreign debt is an order of magnitude larger than than firms who only issue local currency debt, but as Figure 2 shows, smaller firms do access foreign markets although at lower frequency. Operationally, foreign currency issuers exhibit moderately higher real sales growth (1.9% versus 1.4% quarterly). As described in Salomao and Varela (2021), this can be consistent with the fact that firms who are able to borrow in foreign curreny are more productive or that cheaper foreign currency borrowing allows firms to grow faster. However, they maintain lower cash buffers, with lower mean liquidity ratios of 9.6%. This reduced liquidity holding among foreign currency issuers may reflect their superior access to capital markets, reducing the need for precautionary savings, or could indicate that servicing foreign currency debt requires more intensive cash management.

Table 5. Firm Statistics by FX Issuance Status

	Has Never Issued FX (732 Firms)				Has Issued FX (287 Firms)			
Variable	Mean	Median	Std Dev	95th Pct.	Mean	Median	Std Dev	95th Pct.
log(Real Assets)	7.837	7.979	1.603	10.204	8.966	9.092	2.008	11.926
Leverage	0.313	0.296	0.160	0.593	0.336	0.315	0.150	0.601
Real Sales Growth	0.014	0.017	0.207	0.318	0.019	0.022	0.187	0.272
Liquidity	0.107	0.087	0.089	0.273	0.096	0.080	0.075	0.246

4 Empirical Analysis

The estimates show that firms with both high leverage and high foreign currency debt shares respond less to monetary easing than firms with similar leverage and low foreign currency debt shares. I first estimate the level of the investment response across firm types, then compare firms within the same country and quarter to isolate differences associated with their debt composition.

4.1 Total Effect of Monetary Policy on Firm Investment

My baseline local projections estimate both the level of firms' investment responses and how those responses vary with foreign currency exposure and leverage. The specification closely follows Ottonello and Winberry (2020):

$$\begin{aligned}
\Delta_h \log k_{i,c,t+h} = {} & \psi_c + \gamma_{sh} + \beta_{1h}\mathrm{MP}_{c,t} + \beta_{2h}\mathrm{FX}_{i,c,t-1} + \beta_{3h}(\mathrm{FX}_{i,c,t-1} \times \mathrm{Lev}_{i,c,t-1}) \\
& + \beta_{4h}(\mathrm{MP}_{ct} \times \mathrm{FX}_{i,c,t-1}) + \beta_{5h}(\mathrm{MP}_{ct} \times \mathrm{Lev}_{i,c,t-1}) \\
& + \beta_{6h}(\mathrm{MP}_{ct} \times \mathrm{FX}_{i,c,t-1} \times \mathrm{Lev}_{i,c,t-1}) + \mathbf{\Delta}'_{1,h} X_{i,c,t-1} + \mathbf{\Delta}'_{2,h} Y_{c,t-1} + \varepsilon_{i,c,t+h}.
\end{aligned}$$

The dependent variable measures cumulative investment rates over horizons $h = 1$ to 8 quarters. The subscripts (i, c, t) represent firm, country and time indices and s represents sectors. The variables MP, FX, and Lev represent identified monetary policy shocks, foreign currency debt share, and leverage, respectively. The monetary policy shock is multiplied by -1 so that an increase represents a decrease in interest rates. The specification includes the interaction terms between the monetary policy shock and both foreign currency exposure and leverage, as well as a triple interaction capturing their joint effect.

The coefficients of interest are β_4, β_5, and β_6. The coefficient β_4 captures how foreign currency exposure alters investment responses to monetary policy; I expect this to be negative if currency mismatches create balance sheet vulnerabilities. The coefficient β_5 identifies the standard leverage channel of monetary policy. Most importantly, β_6 reveals whether highly leveraged firms with high foreign currency exposure respond more or less to a monetary policy surprise. A negative β_6 would indicate that highly leveraged foreign currency borrowers face amplified balance sheet effects when monetary easing induces depreciation—the increase in local currency debt burdens overwhelms the benefits of lower interest rates, creating a perverse effect where easing actually constrains investment.

I lag firm financial characteristics (foreign currency share and leverage) to mitigate concerns about simultaneity—firms cannot instantaneously adjust their debt structure in response to monetary shocks. In contrast, monetary policy shocks enter contemporaneously as they represent exogenous policy innovations that firms observe and respond to within the quarter.

The vector X is a set of controls that includes firm size (log real assets), real sales growth, liquidity ratios, and leverage. These controls address potential confounding factors. Larger firms may have different investment dynamics and foreign currency access. Sales growth proxies for investment opportunities that could correlate with both currency choices and investment responses. Liquidity buffers affect firms' ability to weather shocks and may substitute for external financing. Leverage captures financial constraints that may independently affect investment responses to monetary policy. Since I am interested in the total effect of a monetary policy shock on firms with varying levels of leverage and FX exposure, I cannot include a country time fixed effect because it would absorb the monetary policy shock. To control for country time specific factors that may influence investment, I include a vector of controls Y that includes lagged GDP growth rates and inflation rates. Section 4.3 includes results for a specification with country-time fixed effects to analyze the differential response of firm investment to monetary policy shocks to show that the results are not driven by omitted country-time variables.

Since I am studying firm investment responses to monetary policy shocks *across* firms with different leverage and FX exposure, I do not include firm fixed effects in the main specification. Including firm fixed effects identifies effects from within-firm variation over time—how the same firm's investment responds differently to monetary shocks as its foreign currency exposure changes. In the pooled specification, β_4 captures both the effect of foreign currency exposure on monetary policy transmission and any systematic differences in how foreign currency borrowers respond to monetary shocks. I include regression results with firm fixed effects in the appendix as a robustness check.

Table 6 presents my baseline results examining how foreign currency exposure and leverage interact to shape firm investment responses to monetary policy shocks. The key finding is the consistently negative and statistically significant coefficient on the triple interaction term, which ranges from -0.026 to -0.128 across horizons and peaks around one year.

This negative triple interaction reveals that firms with both high leverage and substantial foreign currency debt respond less to a monetary easing. Several mechanisms could drive this result. If a monetary easing causes the exchange rate to depreciate, firms may face an immediate increase in the local currency value of their debt service obligations, potentially reducing cash flow available for investment. The higher debt burden could also trigger covenant violations or credit rating concerns, possibly restricting access to financing. For highly leveraged firms, this balance sheet shock may necessitate deleveraging rather than new capital expenditures, as firms might prioritize financial stability over growth opportunities.

The exchange-rate responses in Appendix Figure A3 show that monetary easing is followed by depreciation in several sample countries, consistent with this channel, although the direction and persistence of the response vary across countries.

Table 6. Baseline Results

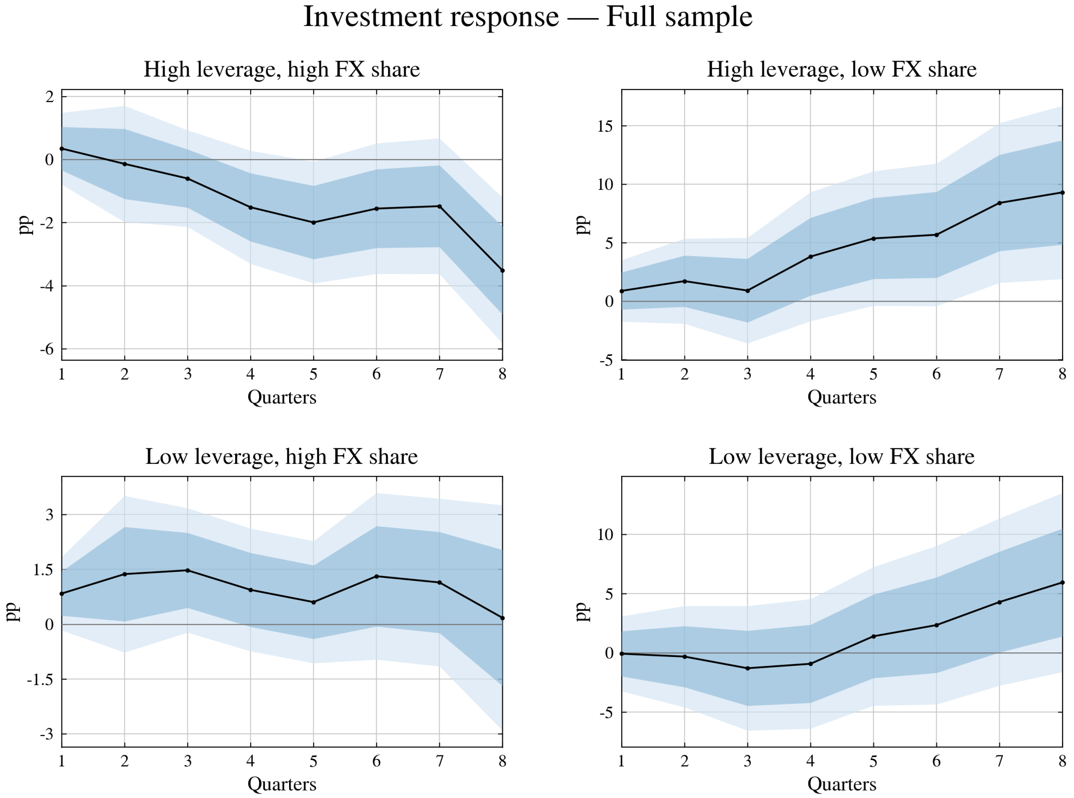
$\Delta_h \log k_{i,c,t+h}$	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8
$\text{MP}_{c,t}$	0.005	0.009	-0.000	0.019	0.037	0.043	0.067*	0.079*
	(0.017)	(0.023)	(0.028)	(0.030)	(0.033)	(0.036)	(0.040)	(0.044)
$\text{FX}_{i,c,t-1}$	0.002	0.005	0.007	0.008	0.008	0.009	0.010	0.010
	(0.005)	(0.007)	(0.007)	(0.007)	(0.009)	(0.009)	(0.010)	(0.009)
$\text{FX}_{i,c,t-1} \times \text{Lev}_{i,c,t-1}$	-0.008*	-0.016***	-0.025***	-0.035***	-0.046***	-0.055***	-0.064***	-0.071***
	(0.004)	(0.006)	(0.007)	(0.008)	(0.009)	(0.010)	(0.011)	(0.012)
$\text{MP}_{c,t} \times \text{FX}_{i,c,t-1}$	0.001	-0.004	0.003	-0.024	-0.047	-0.047	-0.071*	-0.099**
	(0.016)	(0.023)	(0.026)	(0.028)	(0.029)	(0.033)	(0.037)	(0.040)
$\text{MP}_{c,t} \times \text{Lev}_{i,c,t-1}$	0.017	0.036*	0.039	0.084*	0.071	0.060	0.073	0.060
	(0.013)	(0.020)	(0.025)	(0.050)	(0.048)	(0.050)	(0.049)	(0.037)
$\text{MP}_{c,t} \times \text{Lev}_{i,c,t-1} \times \text{FX}_{i,c,t-1}$	-0.026	-0.063***	-0.076***	-0.128**	-0.117**	-0.111*	-0.120**	-0.126**
	(0.016)	(0.024)	(0.029)	(0.057)	(0.057)	(0.060)	(0.059)	(0.053)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector × Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.063	0.079	0.090	0.105	0.112	0.120	0.127	0.144

Standard errors in parentheses
Standard errors clustered by country × time.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The components of the triple interaction reveal distinct channels. The positive coefficient on the leverage and MPS interaction confirms standard monetary transmission: leveraged firms benefit from monetary easing through reduced interest costs and improved refinancing conditions. The direct foreign currency interaction (MP × FX) only becomes statistically significant at late horizons, indicating that foreign currency exposure alone does not significantly alter monetary policy transmission. Only when high leverage combines with foreign currency exposure do the balance sheet vulnerabilities emerge that can overwhelm the traditional interest rate channel.

To more clearly visualize how a monetary easing differentially effects firms with varying levels of leverage and FX, Figure 3 plots the marginal effect of a 1 percentage point monetary easing on investment for a firm with leverage and FX exposure at the 75th and 25th percentiles. The marginal effect implied by my regression specification is given by

$$\frac{\partial \mathbb{E}\Delta_h \log k_{i,c,t+h}}{\partial \text{MP}_{c,t}} = \beta_{1h} + \beta_{4h}\text{FX}_{i,c,t-1} + \beta_{5h}\text{Lev}_{i,c,t-1} + \beta_{6h}(\text{FX}_{i,c,t-1} \times \text{Lev}_{i,c,t-1}).$$


Investment response — Full sample

High leverage, high FX share

High leverage, low FX share

Low leverage, high FX share

Low leverage, low FX share

Figure 3. Marginal effect of a 1pp monetary policy shock. Low leverage and low FX Share are defined as values in the 25th percentile while high leverage and high FX share are defined as values in the 75th percentile. Dark and light blue shaded areas represent 68% and 90% confidence bands, respectively. Responses are in percentage points

The top panels compare the responses of high-leverage firms with high and low foreign currency debt shares. For firms with high leverage and high foreign currency shares, monetary easing generates negative and increasingly adverse effects on capital growth, with the marginal effect reaching approximately -3.5 percentage points by the eighth quarter following a 1 percentage point easing shock. Conversely, firms with high leverage and minimal foreign currency exposure increase investment across all quarters, supporting the idea that higher foreign currency debt exposed firms to balance sheet deterioration. Comparing the top and bottom panels, it is easy to see that firms with higher foreign currency debt exposure increase investment less regardless of leverage.

When examining firms with low foreign currency debt exposure across different leverage levels, my findings present an interesting contrast to Ottonello and Winberry (2020). While they find that low-leverage firms increase investment more than high-leverage firms following monetary policy easing, I observe the opposite pattern: among firms with minimal foreign currency debt, those with higher leverage respond more strongly to monetary policy loosening than their low-leverage counterparts.

I run the same regression on emerging market and developed economies separately to see if the effects of monetary policy are different. Emerging market and developed economies are defined as in table 2. I plot the impulse response to a 1pp monetary policy shock in figure 4 and figure 5 while regression results are reported in Appendix Section A.3. The investment dynamics after a monetary policy shock for EMEs closely follow those in the full sample. The regression coefficients on the monetary policy shock (β_{1h}) are imprecisely estimated, leading to slightly negative effects at shorter horizons. Nevertheless, I find that highly leveraged firms with high FX debt decrease investment, while highly leveraged firms with mostly domestic debt are the most responsive to monetary policy.

Figure 5 plots the dynamics of investment for developed economies. As seen in appendix table, most coefficients are imprecisely estimated. However, the result that firms with high foreign currency borrowing respond less to a monetary easing is true across both high leverage and low leverage firms.

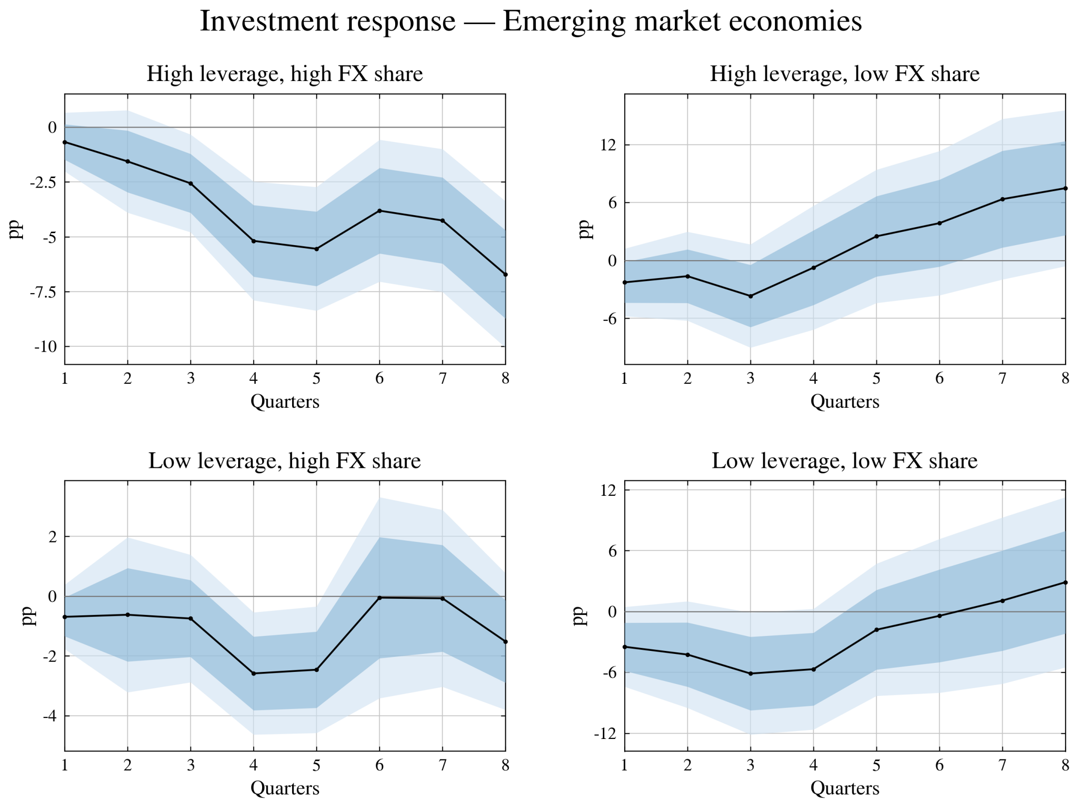


Figure 4. Marginal effect of a 1pp monetary policy shock for emerging market economies. Low leverage and low FX Share are defined as values in the 25th percentile while high leverage and high FX share are defined as values in the 75th percentile. Dark and light blue shaded areas represent 68% and 90% confidence bands, respectively. Responses are in percentage points

Most noticeably, firms in developed economies with high leverage are less responsive to a monetary policy shock conditional on foreign currency debt, as indicated by the negative sign on the coefficent β_{5h} and comparing the top and bottom panels in figure 5. This contrasts with the results for emerging market economies and the full sample, and aligns with the results found in Ottonello and Winberry (2020). Their explanation for this result is that low-leverage firms face a relatively flat cost curve, meaning when monetary policy reduces interest rates, these firms can expand borrowing substantially without hitting steep increases in their cost of capital. High-leverage firms, by contrast, face a steep marginal cost curve due to an increase in default risk.

While the precise reason for the opposite sign responses to monetary policy across high and low leverage firms for developed and EM economies is unclear, one possible reason

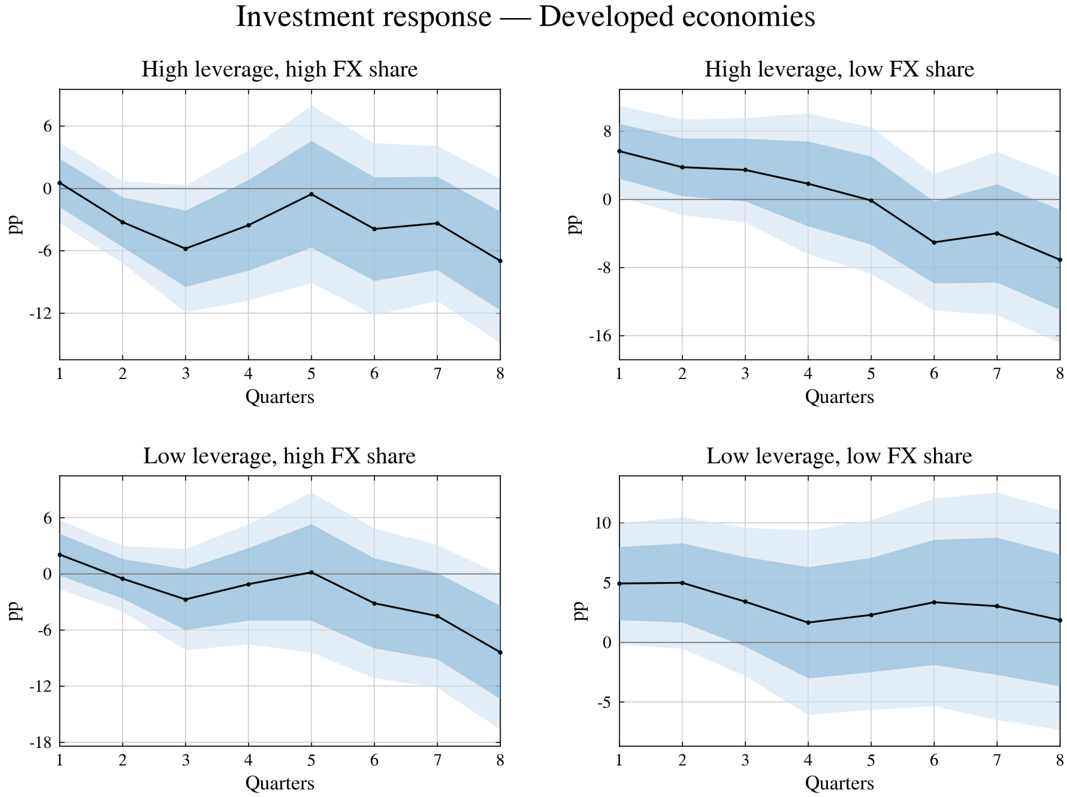


Figure 5. Marginal effect of a 1pp monetary policy shock for developed economies. Low leverage and low FX Share are defined as values in the 25th percentile while high leverage and high FX share are defined as values in the 75th percentile. Dark and light blue shaded areas represent 68% and 90% confidence bands, respectively. Responses are in percentage points

could be that in emerging markets with severe credit rationing, achieving high leverage could signal firm quality—similar to models where debt serves as a costly signal of productivity when information asymmetries are severe—as only the most productive firms can convince lenders to extend substantial credit. In contrast, in developed markets with relatively complete credit access, high leverage primarily signals proximity to financial constraints, consistent with Ottonello-Winberry's findings. These results suggest a potential mechanism: in credit-rationed economies, high leverage may signal firm quality. Testing this hypothesis is a promising avenue for further work.

My results highlight a potential policy tradeoff. While a monetary easing is expansionary for a large subset of firms, it can be immediately contractionary for firms with high leverage and foreign currency debt exposure. These results suggest that aggregate monetary policy effectiveness in emerging markets may depend on the distribution of firm-level financial vulnerabilities, as countries with large shares of highly leveraged, foreign currency-indebted firms may experience counterproductive short-run effects from easing. Furthermore, Table 4 reveals that foreign currency borrowers tend to be larger firms with higher sales growth, indicating that monetary easing's adverse effects are concentrated among firms with significant implications for aggregate growth.

4.2 Tradable and Nontradable Sectors

I do not observe firms' exports or their use of financial hedging instruments, so foreign currency debt shares measure exposure only on the liability side of the balance sheet. Two firms with the same foreign currency debt share can face different currency mismatches if one earns foreign currency revenues. As an indirect proxy for natural hedges, I estimate the baseline specification separately for firms in tradable sectors (agriculture, mining and manufacturing) and nontradable sectors (construction, wholesale and retail trade). Transportation and services are excluded because their broad classifications combine activities with different exposure to international trade.

The weaker response of highly leveraged foreign currency borrowers is concentrated among nontradable firms (Appendix Figures A1 and A2). Following a one percentage point easing, highly leveraged nontradable firms with substantial foreign currency debt reduce cumulative capital growth by about 5.5 percentage points over two years, while their counterparts with little foreign currency debt increase it by about 9.3 percentage points. Among tradable firms, the response of highly leveraged foreign currency borrowers is close to zero and statistically insignificant at every horizon, compared with an increase of about 3.3 percentage points for those with little foreign currency debt. This pattern is consistent with export revenues partially offsetting the rise in the domestic currency value of foreign currency debt after a depreciation. Sector membership does not reveal the currency of export receipts or firms' financial hedges, however, and the two samples use different exposure values and fixed effects, so I interpret the comparison as suggestive rather than as a formal test (see Appendix Section A.4).

4.3 Differential Response to Monetary Policy

The baseline estimates show a weaker investment response among highly leveraged firms with substantial foreign currency debt. I now test whether that difference remains when I compare firms exposed to the same country-quarter conditions. The baseline specification omits country-time fixed effects because they would absorb the monetary policy shock itself.

This section adds a country time fixed effect and removes the monetary policy shock and country level controls. The only variation left for identification comes from within-country, within-quarter differences in the interaction of the policy shock with firm characteristics. I interpret the results in this section as differential effects purely from the cross-section when all firms in the same country and quarter share the same macro shocks. The regression specification is

$$\begin{aligned}\Delta_h \log k_{i,c,t+h} = \psi_{ch} + \gamma_{sh} + \beta_{2h}\mathrm{FX}_{i,c,t-1} + \beta_{3h}(\mathrm{FX}_{i,c,t-1} \times \mathrm{Lev}_{i,c,t-1}) \\ + \beta_{4h}(\mathrm{MP}_{ct} \times \mathrm{FX}_{i,c,t-1}) + \beta_{5h}(\mathrm{MP}_{ct} \times \mathrm{Lev}_{i,c,t-1}) \\ + \beta_{6h}(\mathrm{MP}_{ct} \times \mathrm{FX}_{i,c,t-1} \times \mathrm{Lev}_{i,c,t-1}) + \Gamma_h^{'} X_{i,c,t-1} + \varepsilon_{i,c,t+h}.\end{aligned}$$

Table 7 presents my results. Again, the key finding is the consistently negative and statistically significant coefficient on the triple interaction term, which ranges from -0.026 to -0.125 across horizons and peaks around one year. This negative triple interaction reveals that firms with both high leverage and substantial foreign currency debt respond less to a monetary easing. Interestingly, the level effect of foreign currency share differs from the results in Table 6, with negative and statistically significant effects also most horizons. This suggests that foreign currency borrowers may be inherently different—potentially riskier or more financially constrained firms.

Table 7. Differential Responses to Monetary Policy Shock

$\Delta_h \log k_{i,c,t+h}$	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8
$\text{FX}_{i,c,t-1}$	-0.005	-0.009	-0.014*	-0.019***	-0.020**	-0.022**	-0.024**	-0.028***
	(0.005)	(0.006)	(0.007)	(0.006)	(0.008)	(0.008)	(0.009)	(0.008)
$\text{FX}_{i,c,t-1} \times \text{Lev}_{i,c,t-1}$	-0.004	-0.008	-0.013*	-0.020***	-0.027***	-0.033***	-0.039***	-0.046***
	(0.004)	(0.006)	(0.007)	(0.007)	(0.008)	(0.009)	(0.010)	(0.010)
$\text{MP}_{c,t} \times \text{FX}_{i,c,t-1}$	0.029	0.043	0.037	0.011	0.007	0.026	0.009	-0.042
	(0.018)	(0.030)	(0.032)	(0.022)	(0.025)	(0.036)	(0.038)	(0.029)
$\text{MP}_{c,t} \times \text{Lev}_{i,c,t-1}$	0.013	0.031	0.042*	0.082*	0.065	0.050	0.064	0.035
	(0.013)	(0.021)	(0.025)	(0.049)	(0.046)	(0.051)	(0.045)	(0.035)
$\text{MP}_{c,t} \times \text{Lev}_{i,c,t-1} \times \text{FX}_{i,c,t-1}$	-0.026	-0.069**	-0.088***	-0.125**	-0.115**	-0.117*	-0.122**	-0.100**
	(0.016)	(0.031)	(0.034)	(0.056)	(0.055)	(0.063)	(0.056)	(0.048)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country × Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector × Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.175	0.203	0.217	0.242	0.245	0.247	0.254	0.276

Standard errors in parentheses
Standard errors clustered by country × time.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5 A Model of Foreign Currency Borrowing and Monetary Transmission

This section develops a small open economy model with heterogeneous firms to interpret the evidence in Section 4. The firm block builds on Khan and Thomas (2013), in which firms with decreasing returns to scale invest subject to a collateral constraint and borrow with riskless debt. I extend their framework in two ways. First, firms borrow in both local and foreign currency, with convex issuance costs in each market and a fixed cost of accessing the dollar market, so the currency composition of debt is a choice. Second, I embed the firms in a small open economy with sticky prices and a central bank that sets the policy rate, following Ottonello and Winberry (2020), so that monetary policy moves the interest rate and the exchange rate together.

5.1 Heterogeneous Firms

Production. There is a unit mass of heterogeneous firms owned by domestic households. Firms produce a homogeneous intermediate good with capital and labor, $(k_{i,t}, n_{i,t})$, using a decreasing-returns-to-scale technology

$$y_{i,t} = z_{i,t}^{1-\nu}\, k_{i,t}^{\alpha}\, n_{i,t}^{\nu}, \qquad \alpha + \nu < 1, \tag{1}$$

where idiosyncratic productivity follows $\ln z_{i,t+1} = \rho_z \ln z_{i,t} + \sigma_z \epsilon^z_{i,t+1}$, with $\epsilon^z_{i,t+1}$ a standard normal shock. Firms sell their output to retailers at the nominal price P^m_t and hire labor in a competitive market at the nominal wage W_t. Let $m_t \equiv P^m_t / P_{H,t}$ denote the price of the intermediate good relative to the home good, which is the retailers' real marginal cost (Section 5.3), $p_{H,t} \equiv P_{H,t}/P_t$ the relative price of the home good, and $w_t \equiv W_t/P_t$ the real wage.

Labor is chosen after productivity is realized and does not affect future states, so each period the firm solves a static problem. In units of the household consumption bundle, the firm chooses labor to maximize operating profits given its capital:

$$\pi_{i,t} = \max_{n_{i,t}} \; m_t\, p_{H,t}\, z_{i,t}^{1-\nu}\, k_{i,t}^{\alpha}\, n_{i,t}^{\nu} - w_t\, n_{i,t}. \tag{2}$$

The first-order condition equates the marginal revenue product of labor to the real wage, which gives the labor demand

$$n_{i,t} = \left(\frac{\nu\, m_t\, p_{H,t}}{w_t} \right)^{\frac{1}{1-\nu}} z_{i,t}\, k_{i,t}^{\frac{\alpha}{1-\nu}}. \tag{3}$$

Labor demand is linear in productivity, increasing in capital, and decreasing in the real wage. Substituting (3) into (2), the wage bill is a share ν of revenue, and optimal operating profits are

$$\pi_{i,t} = (1-\nu)\, \nu^{\frac{\nu}{1-\nu}}\, \Theta_t\, z_{i,t}\, k_{i,t}^{\frac{\alpha}{1-\nu}}, \qquad \Theta_t \equiv (m_t\, p_{H,t})^{\frac{1}{1-\nu}}\, w_t^{-\frac{\nu}{1-\nu}}. \tag{4}$$

Capital. Firms accumulate capital subject to adjustment costs,

$$k_{i,t+1} = (1-\delta) k_{i,t} + I_{i,t} - \Psi_k(k_{i,t+1}, k_{i,t}), \tag{5}$$

where $I_{i,t}$ is investment in units of capital goods, $\delta \in (0,1)$ is the depreciation rate, and Ψ_k is a convex adjustment cost, also in units of capital goods,

$$\Psi_k(k_{i,t+1}, k_{i,t}) \equiv \frac{\psi_k}{2} \left(\frac{k_{i,t+1} - (1-\delta)k_{i,t}}{k_{i,t}} \right)^2 k_{i,t}. \tag{6}$$

Firms buy and sell capital goods at the relative price q_t, set by the capital producer (Section 5.2).

Debt. Firms borrow with one-period bonds denominated in local or foreign currency, which I refer to as pesos and dollars. A firm issues $B_{i,t+1} \geq 0$ pesos of face value at the price $1/(1+i_t)$, where i_t is the domestic policy rate, and $B^*_{i,t+1} \geq 0$ dollars of face value at the price $1/(1+i^*)$, where i^* is the foreign interest rate. These prices are set by domestic households, who lend in pesos (Section 5.4), and by foreign investors, who lend in dollars (Section 5.5).[2] Let $\mathcal{E}_t$ denote the nominal exchange rate, the price of one dollar in pesos, P^*_F the dollar price of the foreign good, and $e_t \equiv \mathcal{E}_t P^*_F / P_t$ the real exchange rate, so an increase in e_t is a real depreciation. I work with the real face values $b_{i,t+1} \equiv B_{i,t+1}/P_t$ and $b^*_{i,t+1} \equiv B^*_{i,t+1}/P^*_F$. Issuing $b_{i,t+1}$ raises $b_{i,t+1}/(1+i_t)$ units of the consumption bundle today and requires a real repayment of $b_{i,t+1}/\Pi_{t+1}$ tomorrow, where $\Pi_{t+1} \equiv P_{t+1}/P_t$ is gross CPI inflation. Issuing $b^*_{i,t+1}$ raises $e_t\, b^*_{i,t+1}/(1+i^*)$ today and requires a real repayment of $e_{t+1} b^*_{i,t+1}$ tomorrow.

Cash on hand. After production and the repayment of inherited debt, the firm's real cash on hand is

$$x_{i,t} = \pi_{i,t} + q_t(1-\delta)k_{i,t} - \frac{b_{i,t}}{\Pi_t} - e_t\, b^*_{i,t}. \tag{7}$$

Cash on hand is the sum of operating profits and the market value of undepreciated capital, net of debt repayments. Equation (7) contains the two revaluation channels of monetary policy. Inherited peso debt is a nominal claim, so unexpected inflation (a higher Π_t) lowers its real burden. Inherited dollar debt is repaid at the current real exchange rate, so a depreciation (a higher e_t) raises its real burden.

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[2] This differs from Salomao and Varela (2021) and Hegarty et al. (2024), where foreign currency debt is defaultable and its price falls with the firm's exposure to exchange rate risk. Abstracting from default isolates the revaluation channel of a policy-induced depreciation, which is the channel suggested by the evidence in Section 4. Section 5.9 discusses this choice.

Financing and costly equity. At the end of period t, the firm finances next period's capital with cash on hand, new peso and dollar debt, and outside equity. The cash-flow constraint is

$$q_t\left[k_{i,t+1} + \Psi_k(k_{i,t+1}, k_{i,t})\right] + \mathcal{H}_t(b_{i,t+1}, b^*_{i,t+1}, k_{i,t+1}) + d_{i,t} = x_{i,t} + \frac{b_{i,t+1}}{1+i_t} + \frac{e_t\, b^*_{i,t+1}}{1+i^*}, \tag{8}$$

where $d_{i,t}$ denotes dividends in units of the consumption bundle and $\mathcal{H}$ is the cost of issuing debt, defined below. A negative value of $d_{i,t}$ is an equity injection from shareholders. Raising equity is costly: when $d_{i,t} < 0$, shareholders bear the additional cost

$$\mathcal{C}(d_{i,t}) = \mathbb{1}\{d_{i,t} < 0\}\left(\varphi_1|d_{i,t}| + \frac{\varphi_2}{2}d^2_{i,t}\right), \tag{9}$$

while paying dividends is costless.[3] Costly equity makes internal funds valuable. A firm with ample cash on hand finances investment out of retained earnings and pays the remainder as dividends. A firm with little cash on hand relies on debt, and turns to outside equity only at a premium. As a result, a fall in cash on hand, such as a revaluation loss on dollar debt, lowers the firm's investment whenever it is not paying dividends.

Debt issuance costs. Issuing debt is also costly, and the costs differ by currency. The real cost of issuing $(b_{i,t+1}, b^*_{i,t+1})$ for a firm that chooses capital $k_{i,t+1}$ depends on the real proceeds raised in each market:

$$\begin{gathered}\mathcal{H}_t(b_{i,t+1}, b^*_{i,t+1}, k_{i,t+1}) = \frac{\eta}{2}\left(\frac{b_{i,t+1}}{1+i_t}\right)^2 + \frac{\eta^*(k_{i,t+1})}{2}\left(\frac{e_t\, b^*_{i,t+1}}{1+i^*}\right)^2 + c_F\,\mathbb{1}\{b^*_{i,t+1} > 0\}, \\ \eta^*(k) \equiv \eta^* k^{-b}.\end{gathered} \tag{10}$$

---

[3] The linear and quadratic terms follow the flotation cost specification of Hennessy and Whited (2007), with the quadratic term capturing evidence that the marginal cost of external finance rises with the amount raised (Altınkılıç and Hansen, 2000).

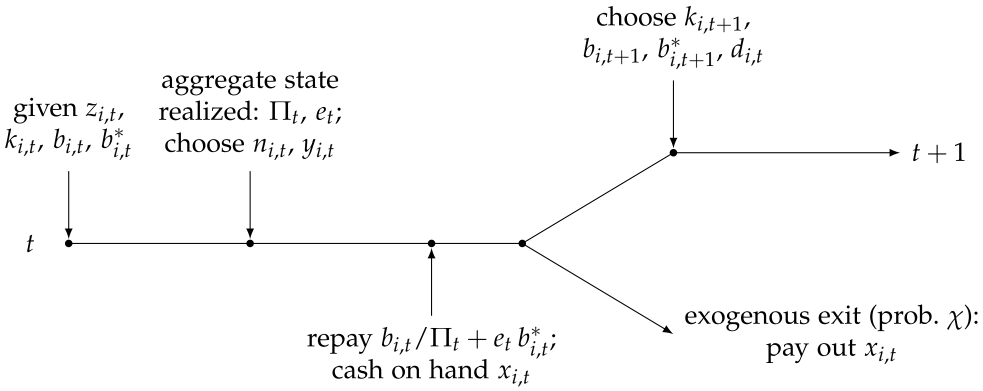
Here η and $\eta^*(k_{i,t+1})$ govern the depth of the peso and dollar markets, and c_F is a fixed cost of accessing the dollar market. For $b > 0$, the dollar market is deeper for larger firms. Section 5.9 discusses this specification.

Collateral constraint. Borrowing is limited by a collateral constraint. The real face value of debt cannot exceed a fraction θ_c of the value of the firm's capital:

$$b_{i,t+1} + e_t\, b^*_{i,t+1} \;\leq\; \theta_c\, q_t\, k_{i,t+1}. \tag{11}$$

Dollar debt enters the constraint at the current exchange rate, so a depreciation tightens the constraint for firms that roll over dollar debt.

Figure 6. Timing


given $z_{i,t}$, $k_{i,t}$, $b_{i,t}$, $b^*_{i,t}$
aggregate state realized: Π_t, e_t; choose $n_{i,t}$, $y_{i,t}$
choose $k_{i,t+1}$, $b_{i,t+1}$, $b^*_{i,t+1}$, $d_{i,t}$
t
$t+1$
repay $b_{i,t}/\Pi_t + e_t\, b^*_{i,t}$; cash on hand $x_{i,t}$
exogenous exit (prob. χ): pay out $x_{i,t}$

Timing. Figure 6 summarizes the timing. A firm enters period t with capital $k_{i,t}$, real debt $(b_{i,t}, b^*_{i,t})$ and productivity $z_{i,t}$. The aggregate state is realized, which determines inflation Π_t, the real exchange rate e_t and the other aggregate prices. The firm then hires labor, produces, and repays its inherited debt at the realized Π_t and e_t, which determines its cash on hand $x_{i,t}$. A monetary policy shock at t therefore revalues debt that was contracted at $t-1$. After production, the firm exits with exogenous probability χ. An exiting firm liquidates its capital, repays its debt, and pays out $x_{i,t}$ to its owners, who bear the equity cost if $x_{i,t} < 0$. Exiting firms are replaced by an equal mass of entrants with capital k_0, no debt, and productivity drawn from the ergodic distribution. A continuing firm chooses $(k_{i,t+1}, b_{i,t+1}, b^*_{i,t+1})$, and its dividend follows from (8).

The firm's problem. The value of a firm solves

$$
\begin{aligned}
V_t(k,b,b^*,z) = \chi\big[x - \mathcal{C}(x)\big] \\
&+ (1-\chi) \max_{k',b',b^{*\prime}} \Big\{ d - \mathcal{C}(d) + \mathbb{E}_t\big[\Lambda_{t,t+1} V_{t+1}(k',b',b^{*\prime},z')\big] \Big\} \\
\text{s.t.} \quad & d = x + \frac{b'}{1+i_t} + \frac{e_t\, b^{*\prime}}{1+i^*} - q_t\big[k' + \Psi_k(k',k)\big] - \mathcal{H}_t(b',b^{*\prime},k'), \\
& b' + e_t\, b^{*\prime} \le \theta_c\, q_t k', \qquad b' \ge 0, \quad b^{*\prime} \ge 0,
\end{aligned} \tag{12}
$$

where cash on hand x is given by (7) and (4), the t subscript encapsulates the aggregate state, primes denote next-period variables, and $\Lambda_{t,t+1}$ is the households' real stochastic discount factor.[4] Firms maximize the expected present value of dividends net of equity issuance costs, discounted with the household's stochastic discount factor because households own all firms.

5.2 Capital Producers

A representative capital producer produces capital goods from home goods subject to adjustment costs. Producing I_t units of capital requires $\Phi(I_t,K_t) = I_t + \frac{\phi}{2}K_t\big(\frac{I_t}{K_t} - \tilde{\delta}\big)^2$ units of the home good, where I_t and K_t denote aggregate investment and capital, and $\tilde{\delta}$ is the steady-state investment rate.[5] Profit maximization implies

$$
q_t = p_{H,t}\left(1 + \phi\Big(\frac{I_t}{K_t} - \tilde{\delta}\Big)\right). \tag{13}
$$

The capital producer is owned by households, to whom its profits are rebated.

---

[4] The constraints $b' \ge 0$ and $b^{*\prime} \ge 0$ in (12) mean that firms cannot accumulate financial savings in either currency. A firm whose cash on hand exceeds its investment needs pays out the excess as dividends rather than holding liquid assets.

[5] Because firms' adjustment costs absorb capital goods even when a firm keeps its capital constant, the steady-state investment rate $\tilde{\delta} \equiv \delta + (1-\chi)\int \Psi_k(k',k)\, d\mu / K$ exceeds δ. Centering the capital producer's cost at $\tilde{\delta}$ makes it zero in steady state, so that $q = p_H$ there.

5.3 Retailers

A continuum of retailers owned by households buys the intermediate good at price P_t^m, differentiates it one for one, and sells varieties of the home good with elasticity of substitution ε_p. Retailers face quadratic price-adjustment costs (Rotemberg, 1982), $\frac{\psi_p}{2}(P_{j,t}/P_{j,t-1}-1)^2 Y_t$ units of the home good. In a symmetric equilibrium, home-good inflation $\Pi_{H,t} \equiv P_{H,t}/P_{H,t-1}$ satisfies

$$(1-\varepsilon_p) + \varepsilon_p\, m_t - \psi_p(\Pi_{H,t}-1)\Pi_{H,t} + \mathbb{E}_t\left[\Lambda_{t,t+1}\frac{p_{H,t+1}Y_{t+1}}{p_{H,t}Y_t}\,\psi_p(\Pi_{H,t+1}-1)\Pi_{H,t+1}\right] = 0. \tag{14}$$

where $\Lambda_{t,t+1}$ is the household's stochastic discount factor (Section 5.4). The intermediate-good price m_t is the retailers' real marginal cost. Its steady-state value is the inverse markup, $(\varepsilon_p - 1)/\varepsilon_p$. CPI inflation combines home-good inflation with changes in the relative price of the home good, $\Pi_t = \Pi_{H,t}\, p_{H,t-1}/p_{H,t}$. A depreciation therefore raises CPI inflation directly through import prices.[6]

5.4 Households

A representative household has preferences

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \psi_l N_t\right] \tag{15}$$

over consumption and labor. Households own all domestic firms, retailers and the capital producer. They trade only peso-denominated assets: equity, firms' peso bonds, and a riskless nominal bond in zero net supply. They cannot borrow or lend in foreign currency.[7] Their budget constraint, in pesos, is

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[6] Linearizing (14) around zero inflation gives $\hat{\Pi}_{H,t} = \frac{\varepsilon_p-1}{\psi_p}\hat{m}_t + \beta\,\mathbb{E}_t\hat{\Pi}_{H,t+1}$. Because labor supply is perfectly elastic and capital is predetermined, $\hat{m}_t = \frac{1-\nu}{\nu}(\hat{Y}_t - \hat{\hat{K}}_t) + \gamma\hat{C}_t + \frac{1-\varpi}{\varpi}\hat{e}_t$. A depreciation thus also acts as a cost-push shock, because wages are paid in units of the consumption bundle.

$$P_t C_t + \frac{\mathcal{B}_{t+1}}{1+i_t} = W_t N_t + \mathcal{B}_t + P_t D_t, \tag{16}$$

where $\mathcal{B}_{t+1}$ denotes the household's holdings of peso bonds, including firms' peso debt, and D_t collects real net cash flows from firms, retailers and the capital producer. From firms, these are dividends net of equity issuance costs, $d_{i,t} - \mathcal{C}(d_{i,t})$, the payouts of exiting firms, and the financing of entrants. The optimality conditions are

$$w_t = \psi_l C_t^{\gamma}, \qquad \frac{1}{1+i_t} = \mathbb{E}_t\left[\frac{\Lambda_{t,t+1}}{\Pi_{t+1}}\right], \qquad \Lambda_{t,t+1} \equiv \beta\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}. \tag{17}$$

The consumption bundle aggregates home and foreign goods with a CES technology,

$$C_t = \left[\varpi^{\frac{1}{\varepsilon}} C_{H,t}^{\frac{\varepsilon-1}{\varepsilon}} + (1-\varpi)^{\frac{1}{\varepsilon}} C_{F,t}^{\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon}{\varepsilon-1}}, \tag{18}$$

where ϖ governs home bias and ε is the trade elasticity. Cost minimization gives the demands $C_{H,t} = \varpi\, p_{H,t}^{-\varepsilon} C_t$ and $C_{F,t} = (1-\varpi)\,(P_{F,t}/P_t)^{-\varepsilon} C_t$, and the consumer price index

$$P_t = \left[\varpi P_{H,t}^{1-\varepsilon} + (1-\varpi) P_{F,t}^{1-\varepsilon}\right]^{\frac{1}{1-\varepsilon}}, \tag{19}$$

where $P_{F,t}$ is the peso price of the foreign good (Section 5.5).

5.5 Rest of the World

The home economy is small relative to the rest of the world, so foreign prices, foreign demand and the foreign interest rate are taken as given.

Prices. The foreign good has a fixed dollar price P_F^* and satisfies the law of one price, $P_{F,t} = \mathcal{E}_t P_F^*$. Because the home economy is small, P_F^* is also the foreign price index. The real exchange rate $e_t = \mathcal{E}_t P_F^* / P_t$ is therefore the relative price of the foreign good in terms of the home consumption bundle, $P_{F,t}/P_t$. As is standard, I normalize $P_F^* = 1$. Under this normalization, a firm's real dollar debt equals its dollar face value, $b_{i,t+1}^* = B_{i,t+1}^*$, and e_t is the real value of one dollar.

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[7] This segmentation follows the assumption in Maggiori et al. (2020) that investors' portfolios are heavily biased toward their own currency, together with the fact that non-US firms' foreign currency borrowing is mostly in dollar bonds held abroad (Section 2). Firms are the only domestic agents that borrow in dollars. Their dollar borrowing therefore makes up the country's net foreign position.

Foreign demand. Home goods are exported at the peso price $P_{H,t}$ (producer currency pricing). Foreign demand for home goods is

$$X_t = \left(\frac{P_{H,t}}{\mathcal{E}_t P_F^*} \right)^{-\varepsilon} C^* = \left(\frac{p_{H,t}}{e_t} \right)^{-\varepsilon} C^*, \tag{20}$$

where C^* denotes foreign absorption. A real depreciation lowers the relative price of home goods abroad and raises exports.

Foreign investors. Foreign investors lend to domestic firms in dollars. They are risk neutral and supply dollar credit perfectly elastically at the nominal rate i_t^*, which follows

$$i_t^* = (1 - \rho^*)\, i^* + \rho^*\, i_{t-1}^* + \varepsilon_t^*,$$

where $1 + i^* = 1/\beta^*$ is the steady-state rate and the innovation ε_t^* can be interpreted as a foreign monetary policy shock. Because firms always repay, a firm's dollar bond is a riskless dollar claim, and its price equals $1/(1 + i_t^*)$, the price used in the firm's problem. Except in Section 6.4.3, $i_t^* = i^*$, and I write i^* for simplicity. Foreign investors do not hold peso assets, so firms' dollar borrowing is the only financial link between the home economy and the rest of the world.

5.6 Monetary Policy

The central bank sets the nominal interest rate according to the Taylor rule $i_t = \rho_i\, i_{t-1} + (1 - \rho_i)\left[\bar{\imath} + \phi_\pi(\Pi_t - 1)\right] + \varepsilon_t^m$, where ρ_i governs interest rate smoothing, ϕ_π is the weight on CPI inflation, $\bar{\imath}$ is the steady-state rate, and ε_t^m is the monetary policy shock.

5.7 Market Clearing and Equilibrium

Distribution of firms. Let μ_t denote the distribution of firms over the idiosyncratic states (k, b, b^*, z) at the beginning of period t, after productivity is drawn and before the exit shock. It has unit mass. All firms in μ_t produce and repay their debt in period t (Figure 6). A fraction χ then exits, and the remaining firms choose $k'_t(\cdot)$, $b'_t(\cdot)$ and $b^{*\prime}_t(\cdot)$, with associated dividends $d_t(\cdot)$. Entrants start with $(k_0, 0, 0)$ and draw productivity from the ergodic distribution. Let $\Gamma_z(\cdot\,|\,z)$ denote the transition probability of productivity and Γ^*_z its ergodic distribution. For any measurable sets $\mathcal{S} \subseteq \mathbb{R}^3_+$ of (k, b, b^*) and $\mathcal{Z}$ of productivity, the distribution evolves according to

$$\begin{aligned} \mu_{t+1}(\mathcal{S} \times \mathcal{Z}) = (1-\chi) \int \mathbb{1}\Big\{ (k'_t, b'_t, b^{*\prime}_t)(k, b, b^*, z) \in \mathcal{S} \Big\}\, \Gamma_z(\mathcal{Z}\,|\,z)\, d\mu_t \\ + \chi\, \mathbb{1}\big\{ (k_0, 0, 0) \in \mathcal{S} \big\}\, \Gamma^*_z(\mathcal{Z}). \end{aligned} \tag{21}$$

The first term moves continuing firms to their chosen capital and debt and draws their next productivity. The second term adds the mass χ of entrants that replaces exiting firms, so μ_{t+1} also has unit mass.

Aggregation. Aggregating across firms,

$$\begin{gathered} \tilde{K}_t \equiv \int z\, k^{\frac{\alpha}{1-\nu}}\, d\mu_t, \qquad K_t \equiv \int k\, d\mu_t, \\ K_{t+1} = (1-\chi) \int k'_t\, d\mu_t + \chi k_0, \qquad I_t = K_{t+1} - (1-\delta) K_t + (1-\chi) \int \Psi_k(k'_t, k)\, d\mu_t, \\ \mathcal{Q}_t \equiv (1-\chi) \int \Big[\mathcal{H}_t(b'_t, b^{*\prime}_t, k'_t) + \mathcal{C}(d_t) \Big] d\mu_t + \chi \int \mathcal{C}(x_t)\, d\mu_t, \end{gathered} \tag{22}$$

where $\tilde{K}_t$ is the productivity-weighted capital stock I_t is gross of the adjustment costs of continuing firms, and $\mathcal{Q}_t$ collects firms' issuance and equity costs, which are paid in units of the consumption bundle. Integrating labor demand (3) across firms gives aggregate output and employment,

$$Y_t = \Big(\frac{\nu\, m_t p_{H,t}}{w_t} \Big)^{\frac{\nu}{1-\nu}} \tilde{K}_t, \qquad N_t = \Big(\frac{\nu\, m_t p_{H,t}}{w_t} \Big)^{\frac{1}{1-\nu}} \tilde{K}_t. \tag{23}$$

Market clearing. Markets clear as follows.

(a) *Labor.* Households supply N_t at the wage $w_t = \psi_l C_t^{\gamma}$.

(b) *Home goods.* Output equals the home-good content of the consumption bundle and of firms' costs, plus the inputs of the capital producer, exports, and the resources used in price adjustment:

$$Y_t = \varpi\, p_{H,t}^{-\varepsilon}\big(C_t + \mathcal{Q}_t\big) + \Phi(I_t, K_t) + X_t + \frac{\psi_p}{2}(\Pi_{H,t} - 1)^2 Y_t. \tag{24}$$

(c) *Capital goods.* The capital producer supplies the capital goods bought by continuing firms, including those absorbed by adjustment costs, and by entrants, net of the undepreciated capital sold by exiting firms: $(1-\chi)\int [k'_t - (1-\delta)k + \Psi_k(k'_t, k)]\, d\mu_t + \chi k_0 - \chi(1-\delta)K_t = I_t$.

(d) *Peso bonds.* Households hold all of firms' peso debt, and the riskless peso bond is in zero net supply: $\mathcal{B}_{t+1} = P_t(1-\chi)\int b'_t\, d\mu_t$.

(e) *Balance of payments.* Households cannot hold foreign assets, so the trade balance finances firms' net dollar debt service. In units of the consumption bundle,

$$\underbrace{p_{H,t}X_t - (1-\varpi)\, e_t^{1-\varepsilon}\big(C_t + \mathcal{Q}_t\big)}_{\text{trade balance}} = \underbrace{e_t \int b^*\, d\mu_t}_{\text{dollar repayments}} - \underbrace{(1-\chi)\int \frac{e_t\, b_t^{*\prime}}{1+i^*}\, d\mu_t}_{\text{new dollar borrowing}}. \tag{25}$$

All firms in μ_t, including those that exit, repay their dollar debt. A shift in firms' borrowing from dollars to pesos lowers new dollar borrowing, and the real exchange rate must depreciate to raise the trade balance. This is how firms' currency choices move the exchange rate.

Definition 1. Given an initial distribution μ_0 and a path of monetary policy shocks $\{\varepsilon_t^m\}_{t\geq 0}$, an equilibrium consists of

(1) prices $\{w_t, m_t, p_{H,t}, q_t, e_t, i_t, \Pi_t, \Pi_{H,t}\}$;

(2) household allocations $\{C_t, N_t, \mathcal{B}_{t+1}\}$;

(3) firm value functions $\{V_t\}$, policies $\{k'_t, b'_t, b^{*\prime}_t\}$ and distributions $\{\mu_t\}$;

(4) aggregates $\{Y_t, \tilde{K}_t, K_t, I_t, X_t, Q_t\}$;

such that, for all t:

(i) given prices, V_t solves (12), $(k'_t, b'_t, b^{*\prime}_t)$ attain the maximum, and $\{\mu_t\}$ evolves according to (21);

(ii) household choices satisfy (17);

(iii) home-good inflation satisfies (14); relative prices satisfy $1 = \varpi p_{H,t}^{1-\varepsilon} + (1-\varpi)e_t^{1-\varepsilon}$ and $\Pi_t = \Pi_{H,t}\, p_{H,t-1}/p_{H,t}$;

(iv) the price of capital satisfies (13), and the policy rate follows the Taylor rule in Section 5.6;

(v) aggregates satisfy (22)–(23), exports satisfy (20), and markets clear as in (a)–(e).

5.8 Model Mechanisms

The currency choice. Let $\lambda_{i,t} \equiv 1 + \mathcal{C}'(d_{i,t})$ denote the firm's marginal value of internal funds. It equals one for a firm paying dividends and exceeds one for a firm raising equity. Consider a firm that borrows in both currencies and whose collateral constraint does not bind. Combining the first-order conditions for $b_{i,t+1}$ and $b^*_{i,t+1}$ in a perfect-foresight path (Appendix B.2) gives

$$1 + i_t = (1 + i^*)\,\frac{\mathcal{E}_{t+1}}{\mathcal{E}_t}\,\frac{1 - \eta\, b_{i,t+1}/(1+i_t)}{1 - \eta^*(k_{i,t+1})e_t\, b^*_{i,t+1}/(1+i^*)}. \tag{26}$$

Equation (26) is an uncovered interest parity condition with a firm-specific wedge that reflects market depth. Each firm equates the expected cost of the two currencies, net of the marginal issuance costs in each market. If both markets were infinitely deep ($\eta, \eta^* \to 0$), the condition would reduce to standard UIP, and the currency composition of debt would be indeterminate. With finite depth, a firm that borrows more spreads its borrowing across both markets to lower its marginal issuance costs. Only firms that borrow enough find it worthwhile to pay the fixed cost c_F, so dollar borrowers are large and fast-growing firms with high borrowing needs. This is the selection pattern documented in Section 3. Once a firm borrows in dollars, its dollar share is pinned down by (26). In a steady state with $i = i^*$, it equals $\eta/(\eta + \eta^*(k_{i,t+1}))$, which rises with the firm's capital for $b > 0$, so larger firms both select into dollar debt and hold a larger share of their debt in dollars. The size dependence also links the capital and currency choices. For a dollar borrower, a larger capital stock lowers the cost of dollar debt at the rate $-\partial \mathcal{H}_t / \partial k_{i,t+1} = \frac{b}{2}\eta^* k_{i,t+1}^{-b-1} \left(e_t b^*_{i,t+1} / (1 + i^*) \right)^2$, which adds to the marginal value of investment.

5.9 Discussion of Baseline Assumptions

First, the model treats the currency composition of debt in reduced form. Convex issuance costs in each currency and a fixed cost of accessing the dollar market let me discipline separately the extensive margin of foreign currency borrowing, which firms borrow in dollars, and the intensive margin, how much of their debt is in dollars, including how that share varies with firm size, while keeping the model tractable. The convex costs capture the limited depth of each market: the marginal cost of borrowing rises with the amount raised in that market, as in the segmented debt markets of Maggiori et al. (2023).[8] For $b > 0$, the marginal cost of a given amount of dollar debt falls with the firm's capital, while the depth of the peso market is the same for all firms. The fixed cost c_F captures the setup costs of accessing foreign currency markets, such as ratings, documentation and underwriting, which are large relative to the debt raised for small issuers. A risk-based motive for currency choice, such as hedging or default risk that varies with the exchange rate, would require aggregate exchange rate risk, which makes the distribution of firms an infinite-dimensional state variable. Because debt is riskless, the revaluation of dollar debt affects investment through the firm's marginal value of internal funds rather than through credit spreads. This choice matches the scope of the evidence: the empirical analysis in Section 4 speaks to how firms with foreign currency exposure respond to monetary policy, not to why they take on that exposure. Salomao and Varela (2021), Maggiori et al. (2020) and Eren et al. (2022) provide explicit reasons for firms to borrow in foreign currency, and embedding a microfounded portfolio decision in this framework is left for future research.

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[8] In Maggiori et al. (2023), investors are segmented by currency, so a firm issuing Q_k in currency k pays the rate $\bar{R} + \Gamma_k Q_k$, and borrowing in foreign currency requires a fixed cost. Borrowing costs are therefore quadratic in the amount raised in each currency. The parameters η, η^* and c_F play the roles of their Γ_k and fixed cost, and in both specifications only firms with large borrowing needs pay the fixed cost. Letting the depth of the dollar market rise with firm size extends their specification, in which Γ_k is common to all firms, and is motivated by their evidence that larger firms select into foreign currency issuance.

Second, firms' revenues are in domestic currency. Their dollar debt is therefore unhedged, apart from the general equilibrium effect of the exchange rate on demand for home goods through exports. The evidence in Section 4.2 suggests that tradable-sector firms are partly hedged by export revenues. The model therefore speaks most directly to the nontradable-sector firms, for which the gap in investment responses by currency of debt is largest.

Finally, debt is one-period and wages are flexible. One-period debt keeps the firm's state low-dimensional and makes all debt due each period exposed to the exchange rate. Long-term debt would spread the revaluation over time. Nominal rigidities enter only through retail prices, so the depreciation passes through to the consumer price index, and through the Taylor rule to the policy rate, via import prices and marginal cost.

6 Quantitative Analysis

This section asks whether the mechanism of Section 5 accounts for the differential investment responses in Section 4, and what it implies for the aggregate effects of monetary policy.

6.1 Calibration

I calibrate the model to the firms in my sample. The calibration targets moments of the cross-section of firms, including their foreign currency borrowing. One period is a quarter. I calibrate the model in two steps.

In the first step, I fix a subset of parameters to standard values in the literature. These are reported in panel A of Table 8. I set the discount factors to $\beta = \beta^* = 0.99$, which implies an annual interest rate of 4 percent in both currencies, and relative risk aversion to $\gamma = 2$. Following Ottonello and Winberry (2020), I set the production elasticities to imply returns to scale of 0.85 and a capital share of one third, the depreciation rate to $\delta = 0.025$, and the exit probability to $\chi = 0.02$, an annual exit rate of 8 percent. I set the elasticity of substitution across retail varieties to $\varepsilon_p = 10$, which implies a markup of 11 percent, and the inflation coefficient in the Taylor rule to $\phi_\pi = 1.1$. The remaining parameters in panel A, which govern trade, price setting, monetary policy and capital production, are set to standard values in the literature.

The second step targets steady-state moments with the parameters in panel B of Table 8. The targets are the moments of the cross-section of firms in Section 3, reported in Table 9, together with the annual frequency of equity issuance from Hennessy and Whited (2007).[9] The collateral parameter θ_c mostly determines leverage, the cost of equity φ_1 the frequency of equity issuance, the productivity process the dispersion of size and sales growth, the capital adjustment cost ψ_k the dispersion of investment, and the capital of entrants k_0 mean investment. The most novel part of the calibration is the cost of issuing debt. The depth of the peso market η and the size elasticity of the dollar market's depth b determine the share of firms that borrow in dollars and how their dollar share varies with size, which together pin down the mean and dispersion of foreign currency shares. The fixed cost c_F determines participation in the dollar debt market. The model matches the foreign currency moments and the size gap closely, but it understates mean leverage and the dispersion of firm size.

6.2 Who Borrows in Dollars

Figure 7 and Table 10 compare firms with and without foreign currency debt in the model and the data. As in the data, firms with dollar debt are larger, by 1.34 log points against 1.13 in the data, and somewhat more levered. Selection follows from the issuance

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[9] In the model, leverage is debt due relative to capital, size is log capital, and firms with foreign currency debt are those that issue dollar debt in the quarter.

Table 8. Parameters

A. Fixed parameters		
$\beta = \beta^*$	Discount factors	0.99
γ	Relative risk aversion	2
$\alpha,\ \nu$	Production elasticities	0.283, 0.567
δ	Depreciation	0.025
χ	Exit probability	0.02
ε_p	Retail elasticity	10
ϕ_π	Taylor rule, inflation	1.1
ρ_i	Taylor rule, smoothing	0.70
ε	Trade elasticity	0.96
ψ_p	Price-adjustment cost	160
ϕ	Capital producer adj. cost	12
ρ_m	Monetary shock, persistence	0.40
σ_m	Monetary shock, size (ann.)	0.25%
ϖ	Home bias	0.757

B. Calibrated parameters		
θ_c	Collateral parameter	0.581
φ_1	Linear cost of equity	1.030
φ_2	Quadratic cost of equity	0.021
ψ_k	Capital adjustment cost	0.657
ρ_z	Productivity, persistence	0.948
σ_z	Productivity, std. dev.	0.228
k_0	Capital of entrants	$0.23\,\bar{K}$
η	Depth of peso market	3.1×10^{-4}
η^*	Depth of dollar market	2.6×10^{-3}
b	Size elasticity of dollar depth	1.02
c_F	Fixed cost of dollar debt	0.072

Note. Costs are in units of the consumption bundle, in which mean capital is $\bar{K} = 46.4$ and mean quarterly earnings are 3.19.

Table 9. Targeted moments

Moment	Data	Model
Mean leverage	0.321	0.249
SD of leverage	0.157	0.202
Share of firms with foreign currency debt	0.282	0.330
Mean foreign currency share of debt	0.259	0.262
SD of foreign currency share of debt	0.398	0.392
Share of firms issuing equity	0.099	0.103
Mean investment	0.010	0.011
SD of investment	0.169	0.210
SD of sales growth	0.244	0.261
SD of log size	1.831	1.422
Log size, FX borrowers minus others	1.129	1.213

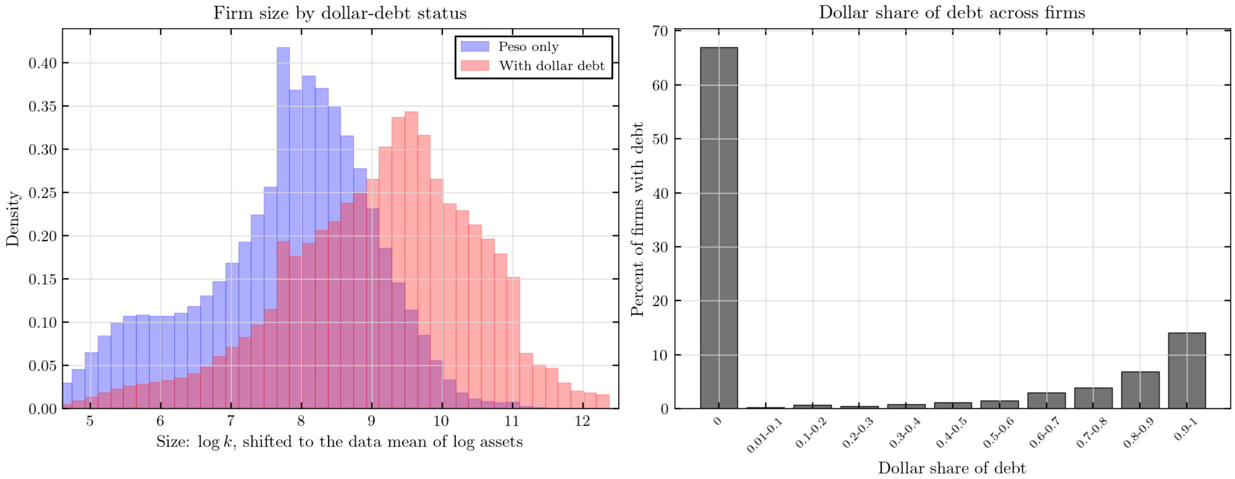

Figure 7. Dollar debt in the stationary equilibrium. Left panel: distribution of firm size among firms that borrow only in pesos and firms with dollar debt. Size is log capital, shifted so that its mean equals mean log assets in the data. Right panel: distribution of the dollar share of debt across firms with debt.

costs: a firm pays the fixed cost c_F only if its borrowing needs are large, and because the dollar market is deeper for larger firms, larger firms also hold a larger share of their debt in dollars. As a result, the distribution of dollar shares is bimodal. Two-thirds of firms with debt borrow only in pesos, while most dollar borrowers hold more than half of their debt in dollars. The model does not match sales growth, which is lower for dollar borrowers in the model and higher in the data.

Table 10. Firms with and without foreign currency debt

	Data		Model	
	No FX debt	FX debt	No FX debt	FX debt
Share of firms	0.72	0.28	0.67	0.33
Log size (relative)	0.00	1.13	0.00	1.34
Leverage	0.31	0.34	0.30	0.36
Sales growth (quarterly)	0.014	0.019	0.050	0.016

Note. Data from Table 5. Log size is relative to firms without FX debt. In the model, firms with foreign currency debt are firms with debt that issue dollar debt in the quarter, size is log capital, leverage is debt due relative to capital, and sales growth is expected quarterly output growth.

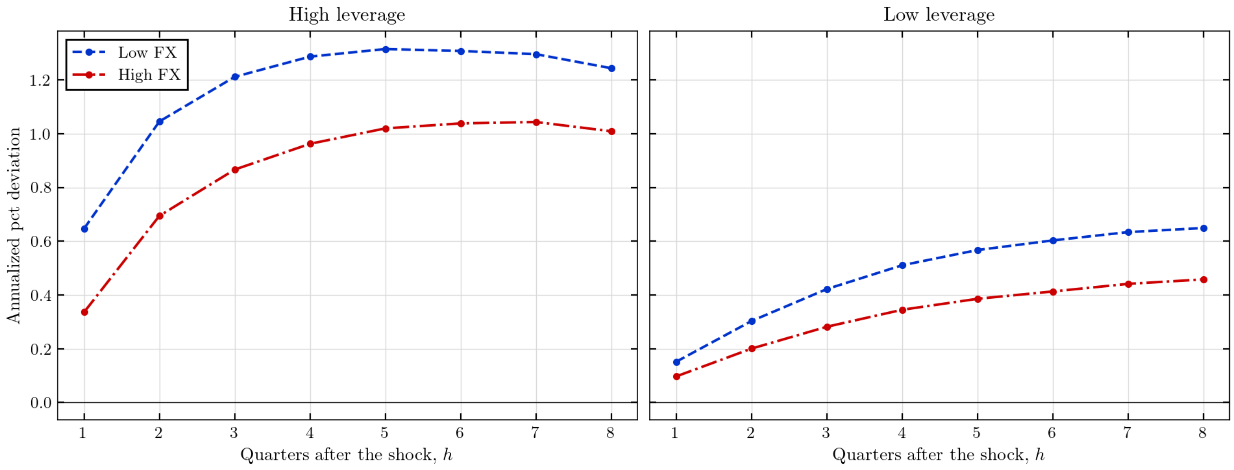

Figure 8. Model-implied investment responses to a monetary easing, by leverage and FX share. Fitted values from a regression of firm-level responses $\Delta_h \log k$ on FX share, leverage and their interaction across firms with debt in the stationary distribution, evaluated at high leverage ($R/k = 0.521$) and low leverage ($R/k = 0.227$), and at low and high FX shares. Responses are in percentage points per 1 pp easing shock at an annual rate.

6.3 Differential Responses

Model counterpart of the empirical estimates. The model counterpart of the shocks in Section 4 is an unanticipated easing shock to the Taylor rule, $\varepsilon_t^m = -\sigma_m \rho_m^t$, with size $\sigma_m = 0.25$ percentage points (annualized) and persistence $\rho_m = 0.4$ (Table 8). Following Ottonello and Winberry (2020, Section 5.2), I compute each firm's response in the stationary distribution, $y_{h,i} = 100\big[\log \mathbb{E}(k_{t+h} \mid i, \text{shock}) - \log \mathbb{E}(k_{t+h} \mid i, \text{no shock})\big]$, exactly and without simulation. I then regress $y_{h,i}$ on FX share, leverage and their interaction across firms with debt, weighted by the stationary distribution. With the shock normalized to 1 percentage point at an annual rate, the coefficients correspond to the interactions with the shock in Table 6. Figure 8, the counterpart of Figure 3, plots fitted responses for high leverage ($R/k = 0.521$) and low leverage ($R/k = 0.227$), at low and high FX shares.

The model reproduces the empirical pattern. Highly levered firms respond more than less levered firms at every horizon, and a high FX share lowers the response, more so for highly levered firms. After four quarters, highly levered firms with a low FX share raise their capital stock by 1.3 percentage points, against 1.0 for those with a high FX share. Among less levered firms, the corresponding responses are 0.5 and 0.3 percentage points. The relevant object for the triple interaction β_6 is the vertical distance between the low-FX (blue) and high-FX (red) lines in each panel, which measures how much a high FX share lowers the response at a given level of leverage. A negative β_6 means that this gap is larger for highly levered firms. In Figure 8, the gap is larger in the high-leverage panel at every horizon, about 0.3 percentage points against at most 0.2 among less levered firms, consistent with $\beta_6 < 0$ in Table 6. The higher level of both lines in the high-leverage panel is consistent with $\beta_5 > 0$.

Leverage raises the response, as in the full sample and among emerging markets in Section 4 and unlike in Ottonello and Winberry (2020). Without default risk, leverage does not steepen the marginal cost of borrowing. Highly levered firms are instead close to their collateral constraint, which the easing relaxes: a lower i_t raises the proceeds each unit of collateral supports, and a higher price of capital raises the value of collateral. Because these firms value internal funds above one, they invest the added borrowing capacity. A high FX share lowers the response for two reasons. First, the surprise depreciation raises the peso value of dollar debt due by $\text{Lev} \times \text{FX} \times \Delta \ln e$ (Appendix B.3), a loss borne by the firms that value internal funds most.

Where the model falls short. The model weakens the response of highly levered dollar borrowers but does not generate the contraction in the data, where their capital stock falls by about 3.5 percentage points after eight quarters. Three features limit the balance-sheet effect. Debt is one-period, so the depreciation revalues only the repayment due at the shock. The expected appreciation that follows lowers the cost of new dollar borrowing. The gains from lower rates and a looser collateral constraint persist, while the revaluation loss occurs once. Long-term dollar debt, default risk and a tighter collateral constraint would each strengthen the mechanism.

6.4 Aggregate Outcomes

6.4.1 Baseline Response to a Monetary Easing

Figure 9 shows the aggregate response to a monetary easing, a 0.25 percentage point (annualized) shock to the Taylor rule with persistence $\rho_m = 0.4$. The nominal policy rate falls by only 0.05 to 0.08 percentage points at an annual rate, because the rule responds to the rise in inflation, but the real rate falls by about 0.3 percentage points. The peso depreciates by 0.2 percent in real terms on impact before appreciating gradually. CPI inflation rises by 0.8 percentage points on impact, as the depreciation raises import prices, and returns to steady state within about six quarters. Investment rises by 0.8 percent, exports by 0.3 percent and consumption by 0.1 percent. Output rises by 0.3 percent, while imports fall by 0.1 percent as the depreciation switches expenditure toward home goods.

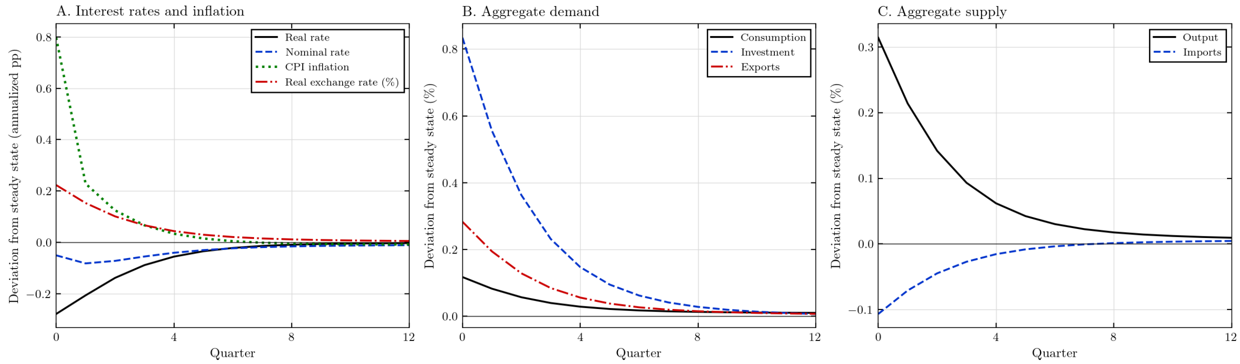

Figure 9. Aggregate responses to a monetary easing. The shock is a 0.25 pp (annualized) easing shock to the Taylor rule with persistence $\rho_m = 0.4$. Panel A: real and nominal policy rates and CPI inflation in annualized percentage points, and the real exchange rate in percent, where an increase is a depreciation. Panels B and C: percent deviations from the stationary equilibrium.

6.4.2 No Dollar Debt

To isolate the role of foreign currency borrowing, I consider a policy that prohibits dollar borrowing, $c_F \to \infty$, and recompute the stationary equilibrium. All other parameters, the Taylor rule and the shock are unchanged. Because households cannot hold foreign assets, the economy is then in financial autarky: the balance of payments (25) requires balanced trade every period, and no firm's choices satisfy the interest parity condition (26). Figure 10 compares the responses with the baseline. Without dollar debt, investment rises by 0.95 percent on impact, against 0.83 percent in the baseline, in part because no firm bears a revaluation loss on dollar debt, and it remains higher for about two years. The peso depreciates by less, 0.19 percent against 0.22 percent, and the depreciation dies out faster. With dollar debt, the depreciation raises the value of dollar repayments due, so the balance of payments requires a larger trade surplus and hence a larger depreciation. In financial autarky, the exchange rate moves only to balance trade. Output, the real interest rate, inflation and consumption respond almost identically in the two economies. Dollar debt therefore lowers the investment response to an easing by about an eighth on impact, while amplifying the depreciation that the easing generates.

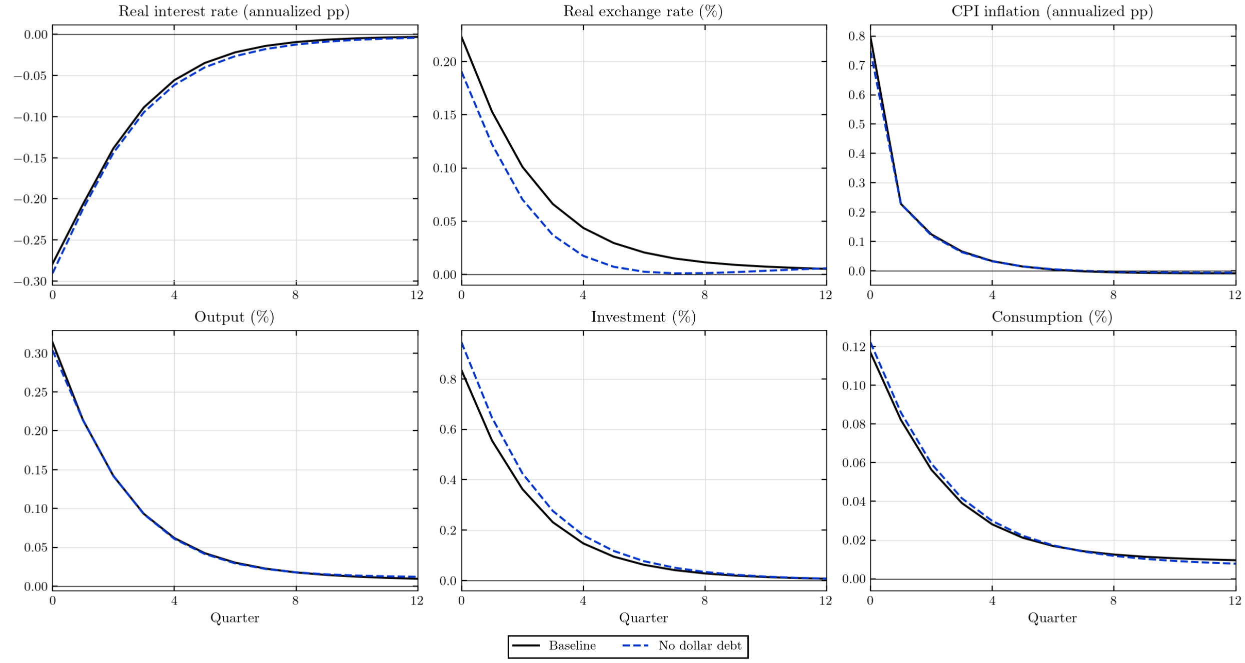

Figure 10. Aggregate responses to a monetary easing with and without dollar debt. The no-dollar-debt economy sets the fixed cost of dollar borrowing to $c_F \to \infty$ and recomputes the stationary equilibrium. Both economies are hit by the same 0.25 pp (annualized) easing shock to the Taylor rule. Interest rates and inflation are in annualized percentage points, and the other variables are in percent deviations from the stationary equilibrium.

6.4.3 Foreign Monetary Policy Shocks and the Monetary Policy Regime

The second exercise asks how the monetary policy rule shapes the transmission of an external shock when firms borrow in dollars. I consider a foreign monetary policy tightening, an unanticipated 25 basis point (annualized) innovation to ε_t^* with persistence $\rho^* = 0.6$, and compare four policy regimes. The three Taylor rules share the smoothing and inflation coefficients of the baseline and take the form

$$i_t = \rho_i \, i_{t-1} + (1-\rho_i)\Big[\bar{\imath} + \phi_\pi \, \pi_t^{\text{target}} + \phi_e \Big(\frac{\mathcal{E}_t}{\mathcal{E}_{t-1}} - 1\Big)\Big].$$

The baseline rule targets CPI inflation, $\pi_t^{\text{target}} = \Pi_t - 1$, with $\phi_e = 0$. The second targets home-goods inflation, $\pi_t^{\text{target}} = \Pi_{H,t} - 1$, with $\phi_e = 0$. The third targets CPI inflation and also responds to depreciation, with $\phi_e = 0.5$. The fourth regime replaces the rule with a nominal exchange rate peg, $\mathcal{E}_t = \mathcal{E}_{t-1}$.

Because households cannot hold foreign assets, the shock reaches the economy only through firms' dollar borrowing. A higher dollar rate makes dollar debt more expensive, less dollar credit flows in, and the balance of payments requires the peso to depreciate. The peso jumps on impact and then appreciates, and the expected appreciation offsets most of the higher dollar rate. The depreciation raises the peso value of dollar debt, raises exports and raises import prices. The policy rule determines how much of the adjustment occurs through the exchange rate and how much through domestic interest rates.

Figure 11 shows the results. Under the two rules that respond only to inflation, the peso depreciates by about 0.13 percent and the policy rate barely moves. Output rises slightly, by about 0.04 percent, because higher exports outweigh the balance-sheet losses of dollar borrowers, while investment and consumption fall only slightly. Leaning against the depreciation raises the policy rate by about 10 basis points. The peso depreciates slightly less, but investment falls by more, 0.10 percent on impact. Under the peg, the central bank must raise the domestic rate one for one with the foreign rate, which keeps the relative cost of peso and dollar debt unchanged and prevents the nominal depreciation. The real exchange rate still moves slightly, through domestic prices. The cost is a domestic contraction: investment falls by 0.41 percent, output by 0.11 percent and consumption by 0.08 percent on impact. The more the central bank leans against the depreciation, the larger the contraction. In this calibration, the balance-sheet cost of a depreciation is small relative to its boost to exports, so defending the currency is costly.

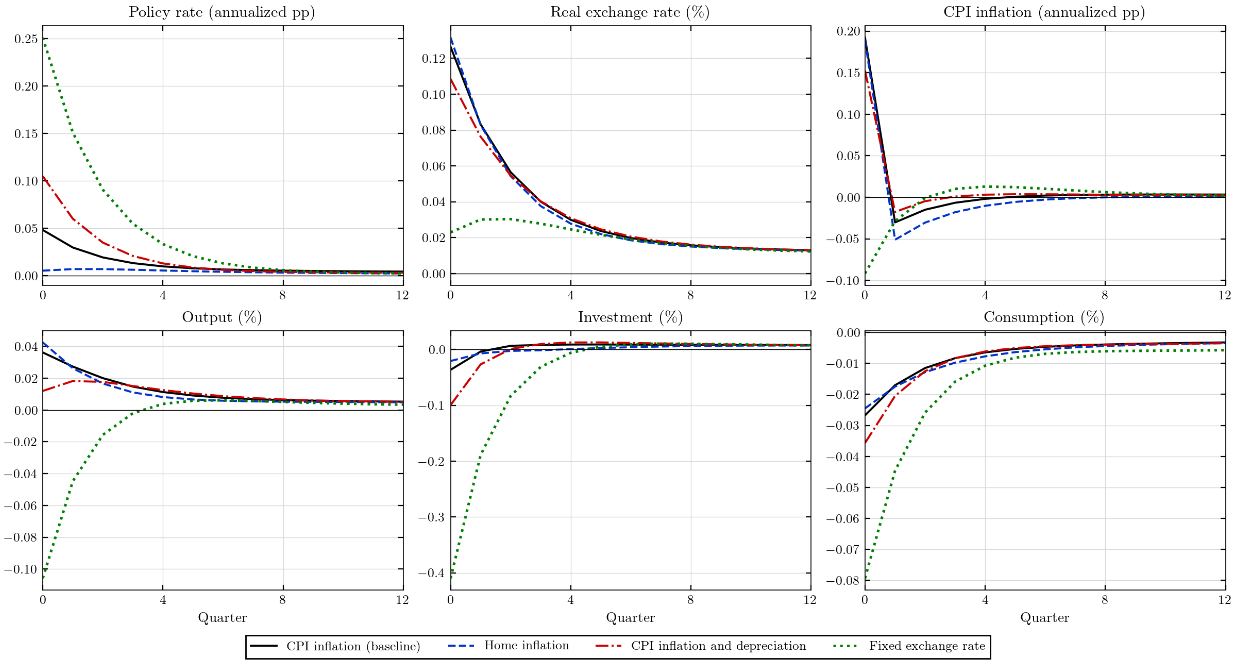

Figure 11. Aggregate responses to a foreign monetary policy tightening, a 25 bp (annualized) innovation to i_t^* with persistence $\rho^* = 0.6$, under alternative monetary policy regimes. All Taylor rules share ρ_i and ϕ_π. “Home inflation” targets home-goods inflation, “CPI inflation and depreciation” adds a response to depreciation with $\phi_e = 0.5$, and “Fixed exchange rate” is a nominal exchange rate peg that holds $\mathcal{E}_t$ constant, so the real exchange rate moves only through prices. The policy rate and inflation are in annualized percentage points, and the other variables are in percent deviations from the stationary equilibrium.

7 Conclusion

Highly leveraged firms with substantial foreign currency debt respond less to monetary easing than firms with similar leverage and less foreign currency debt. Using detailed firm-level data across 15 countries, I find that the predicted investment response for highly exposed firms can be negative, creating a tension in monetary policy transmission. In the baseline estimates, a one percentage point monetary easing predicts an approximately 3.5 percentage point decline in investment after eight quarters for a firm at the 75th percentiles of leverage and foreign currency debt share. The predicted response for a high-leverage firm with low foreign currency debt share is positive.

While firms with foreign currency debt respond less to monetary policy across both developed and emerging market countries, firms in developed countries with higher leverage respond less conditional on foreign currency borrowing. This supports previous findings in the literature that mainly focus on U.S firms.

To interpret these findings, I develop a small open economy model with heterogeneous firms that choose how much to borrow and in which currency. Larger firms select into dollar debt because accessing the dollar market requires a fixed cost, as in the data. A monetary easing depreciates the peso, which raises the real value of dollar debt due and lowers firms' internal funds. For firms near their collateral constraint, investment is a multiple of internal funds, so the loss falls hardest on highly levered dollar borrowers. The model reproduces the signs of the empirical estimates: leverage raises the investment response, and a high foreign currency share lowers it, more so for highly levered firms. In the aggregate, dollar debt lowers the investment response to an easing by about an eighth on impact, while amplifying the depreciation, because the economy must finance higher dollar repayments through the trade balance. Output, consumption and inflation respond almost identically with and without dollar debt. Foreign currency debt therefore shapes which firms benefit from monetary easing more than how much the economy as a whole responds.

The model also speaks to how monetary policy should respond to external shocks. After a foreign monetary policy tightening, dollar credit becomes more expensive and the peso depreciates. Under rules that target inflation, the policy rate barely moves and output rises slightly, because higher exports outweigh the balance-sheet losses of dollar borrowers. Leaning against the depreciation raises the policy rate and deepens the fall in investment, and a nominal exchange rate peg, which requires raising the domestic rate one for one with the foreign rate, produces a domestic contraction, with investment falling by 0.41 percent and output by 0.11 percent on impact. In this calibration, the balance-sheet cost of a depreciation is small relative to its benefits, so defending the currency is costly.

The findings contribute to the understanding of monetary policy transmission in open economies by identifying currency composition as a critical dimension of firm heterogeneity. Rather than treating exchange rate movements as uniformly expansionary through competitiveness gains, policymakers should consider the distribution of foreign currency exposure across leveraged firms when evaluating the effects of policy.

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A Appendix Tables and Figures

A.1 Within-Firm Variation

Appendix Table A1. Baseline Results with Firm Fixed Effects

$\Delta_h \log k_{i,c,t+h}$	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8
$\text{MP}_{c,t}$	0.002	0.004	-0.008	0.007	0.019	0.024	0.051	0.058*
	(0.016)	(0.020)	(0.023)	(0.026)	(0.026)	(0.029)	(0.031)	(0.034)
$\text{FX}_{i,c,t-1}$	0.009	0.015	0.018	0.014	0.016	0.014	0.014	0.011
	(0.009)	(0.012)	(0.013)	(0.013)	(0.014)	(0.015)	(0.016)	(0.016)
$\text{FX}_{i,c,t-1} \times \text{Lev}_{i,c,t-1}$	-0.000	0.000	-0.004	-0.014	-0.014	-0.006	-0.007	0.006
	(0.010)	(0.013)	(0.015)	(0.015)	(0.017)	(0.018)	(0.020)	(0.019)
$\text{MP}_{c,t} \times \text{FX}_{i,c,t-1}$	0.003	-0.002	0.007	-0.015	-0.031	-0.030	-0.057*	-0.080**
	(0.016)	(0.021)	(0.022)	(0.026)	(0.026)	(0.030)	(0.031)	(0.032)
$\text{MP}_{c,t} \times \text{Lev}_{i,c,t-1}$	0.016	0.034**	0.034*	0.084*	0.072	0.061	0.080*	0.065*
	(0.013)	(0.017)	(0.020)	(0.049)	(0.049)	(0.054)	(0.048)	(0.037)
$\text{MP}_{c,t} \times \text{Lev}_{i,c,t-1} \times \text{FX}_{i,c,t-1}$	-0.022	-0.054**	-0.062**	-0.117**	-0.105*	-0.096	-0.104*	-0.101**
	(0.015)	(0.021)	(0.025)	(0.055)	(0.056)	(0.062)	(0.057)	(0.047)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.071	0.131	0.197	0.256	0.288	0.329	0.375	0.439

Standard errors in parentheses
Standard errors clustered by country × time.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The inclusion of firm fixed effects in my baseline specification provides important robustness evidence for my heterogeneity findings. While the average monetary policy effect becomes smaller and more delayed—significant only at horizon 8 rather than horizons 6-7—the key interaction patterns remain qualitatively similar. The triple interaction maintains its negative sign and achieves statistical significance at horizons 4-5 and 7-8, though with somewhat smaller magnitudes than the baseline specification. The monetary policy-leverage interaction continues to exhibit positive and significant effects at early-to-middle horizons, while the foreign exchange interaction turns significantly negative at longer horizons. The preservation of these interaction patterns after controlling for all time-invariant firm characteristics suggests that my heterogeneity results reflect genuine differential responses to monetary policy rather than spurious correlations driven by unobserved firm-level factors.

A.2 Heterogenous responses to monetary policy: Within-Firm variation

This section provides a robustness check for the regression specification in Section 4.3 that exploits within firm variation.

$$\begin{aligned}\Delta_h \log k_{i,c,t+h} = \alpha_i + \psi_{ct} + \gamma_{st} + \beta_2 \mathrm{FX}_{i,c,t-1} + \beta_3(\mathrm{FX}_{i,c,t-1} \times \mathrm{Lev}_{i,c,t-1}) \\ + \beta_4(\mathrm{MP}_{ct} \times \mathrm{FX}_{i,c,t-1}) + \beta_5(\mathrm{MP}_{ct} \times \mathrm{Lev}_{i,c,t-1}) \\ + \beta_6(\mathrm{MP}_{ct} \times \mathrm{FX}_{i,c,t-1} \times \mathrm{Lev}_{i,c,t-1}) + \mathbf{ffi}' X_{i,c,t-1} + \varepsilon_{i,c,t+h}.\end{aligned}$$

Tables A2 present the results examining how foreign currency exposure and leverage interact to shape firm investment responses to monetary policy shocks. Results are largely the same except that the level effects of FX exposure and the interaction between FX and leverage are statistically insignificant. I conjecture that this is due to the peristence in FX exposure within firms, which limits the amount the within firm variation in FX exposure.

Appendix Table A2. Differential Responses with Firm Fixed Effects

	(1) h = 1	(2) h = 2	(3) h = 3	(4) h = 4	(5) h = 5	(6) h = 6	(7) h = 7	(8) h = 8
foreign_share_l	-0.000	0.005	0.013	0.010	0.019	0.027*	0.034**	0.034**
	(0.009)	(0.011)	(0.012)	(0.012)	(0.015)	(0.016)	(0.016)	(0.016)
fx_lev	0.004	0.010	0.011	0.003	0.005	0.013	0.013	0.019
	(0.010)	(0.013)	(0.015)	(0.013)	(0.016)	(0.018)	(0.019)	(0.018)
MPS_fx	0.029	0.043	0.037	0.015	0.016	0.039	0.019	-0.026
	(0.018)	(0.031)	(0.030)	(0.021)	(0.023)	(0.041)	(0.037)	(0.023)
MPS_lev	0.011	0.028	0.036	0.080	0.068	0.058	0.080	0.050
	(0.013)	(0.019)	(0.022)	(0.051)	(0.051)	(0.057)	(0.050)	(0.039)
MPS_lev_fx	-0.023	-0.064**	-0.079**	-0.123**	-0.116**	-0.123*	-0.135**	-0.106**
	(0.015)	(0.032)	(0.032)	(0.057)	(0.058)	(0.069)	(0.060)	(0.046)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country × Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector × Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.179	0.241	0.299	0.360	0.385	0.419	0.460	0.523

Standard errors in parentheses

Standard errors clustered by country × time.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.3 Effects of Monetary Policy: Developed vs Emerging Market Economies

Appendix Table A3. Baseline Results for Developing Economies

	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8
$\text{MP}_{c,t}$	0.053*	0.043	0.034	0.017	0.009	-0.015	-0.011	-0.034
	(0.030)	(0.032)	(0.035)	(0.046)	(0.048)	(0.046)	(0.055)	(0.053)
$\text{FX}_{i,c,t-1}$	0.001	-0.002	-0.004	-0.008	-0.010*	-0.011*	-0.013*	-0.015**
	(0.004)	(0.004)	(0.005)	(0.006)	(0.005)	(0.006)	(0.007)	(0.007)
$\text{FX}_{i,c,t-1} \times \text{Lev}_{i,c,t-1}$	-0.005	-0.012*	-0.017**	-0.023**	-0.033***	-0.038***	-0.045***	-0.052***
	(0.005)	(0.007)	(0.008)	(0.010)	(0.011)	(0.010)	(0.012)	(0.013)
$\text{MP}_{c,t} \times \text{FX}_{i,c,t-1}$	-0.042**	-0.063**	-0.079***	-0.043	-0.011	-0.019	-0.026	-0.040
	(0.020)	(0.029)	(0.030)	(0.031)	(0.036)	(0.041)	(0.045)	(0.046)
$\text{MP}_{c,t} \times \text{Lev}_{i,c,t-1}$	0.013	-0.021	0.001	0.003	-0.043	-0.148**	-0.124**	-0.158*
	(0.034)	(0.038)	(0.050)	(0.055)	(0.057)	(0.070)	(0.060)	(0.083)
$\text{MP}_{c,t} \times \text{Lev}_{i,c,t-1} \times \text{FX}_{i,c,t-1}$	-0.039	-0.028	-0.055	-0.046	0.028	0.127*	0.137**	0.174**
	(0.041)	(0.046)	(0.064)	(0.065)	(0.074)	(0.075)	(0.062)	(0.082)
Constant	0.004	0.005	0.009	0.009	0.007	0.005	0.001	-0.003
	(0.007)	(0.010)	(0.012)	(0.015)	(0.017)	(0.019)	(0.020)	(0.022)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector × Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.158	0.185	0.177	0.179	0.201	0.211	0.217	0.235

Standard errors in parentheses

Standard errors clustered by country × time.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Appendix Table A4. Baseline Results for Emerging Market Economies

	h = 1	h = 2	h = 3	h = 4	h = 5	h = 6	h = 7	h = 8
$\text{MP}_{c,t}$	-0.030	-0.033	-0.052	-0.039	-0.002	0.011	0.030	0.046
	(0.023)	(0.030)	(0.034)	(0.034)	(0.038)	(0.043)	(0.048)	(0.049)
$\text{FX}_{i,c,t-1}$	-0.008	-0.013	-0.021	-0.025*	-0.029	-0.036	-0.041*	-0.048**
	(0.015)	(0.017)	(0.019)	(0.014)	(0.021)	(0.022)	(0.024)	(0.020)
$\text{FX}_{i,c,t-1} \times \text{Lev}_{i,c,t-1}$	0.002	0.004	0.006	0.001	-0.004	-0.006	-0.010	-0.006
	(0.011)	(0.015)	(0.017)	(0.019)	(0.021)	(0.023)	(0.024)	(0.025)
$\text{MP}_{c,t} \times \text{FX}_{i,c,t-1}$	0.023	0.023	0.038	0.004	-0.034	-0.026	-0.046	-0.080*
	(0.021)	(0.029)	(0.033)	(0.032)	(0.035)	(0.043)	(0.045)	(0.043)
$\text{MP}_{c,t} \times \text{Lev}_{i,c,t-1}$	0.022	0.048**	0.044*	0.090*	0.078	0.078	0.096*	0.084*
	(0.014)	(0.023)	(0.026)	(0.053)	(0.051)	(0.055)	(0.056)	(0.044)
$\text{MP}_{c,t} \times \text{Lev}_{i,c,t-1} \times \text{FX}_{i,c,t-1}$	-0.022	-0.065**	-0.077**	-0.137**	-0.134**	-0.146**	-0.172**	-0.178***
	(0.017)	(0.029)	(0.032)	(0.064)	(0.064)	(0.070)	(0.073)	(0.059)
Constant	-0.011	-0.019	-0.019	-0.043	-0.069	-0.092**	-0.092*	-0.083
	(0.022)	(0.029)	(0.035)	(0.037)	(0.043)	(0.046)	(0.048)	(0.051)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector × Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.067	0.086	0.104	0.128	0.128	0.134	0.142	0.161

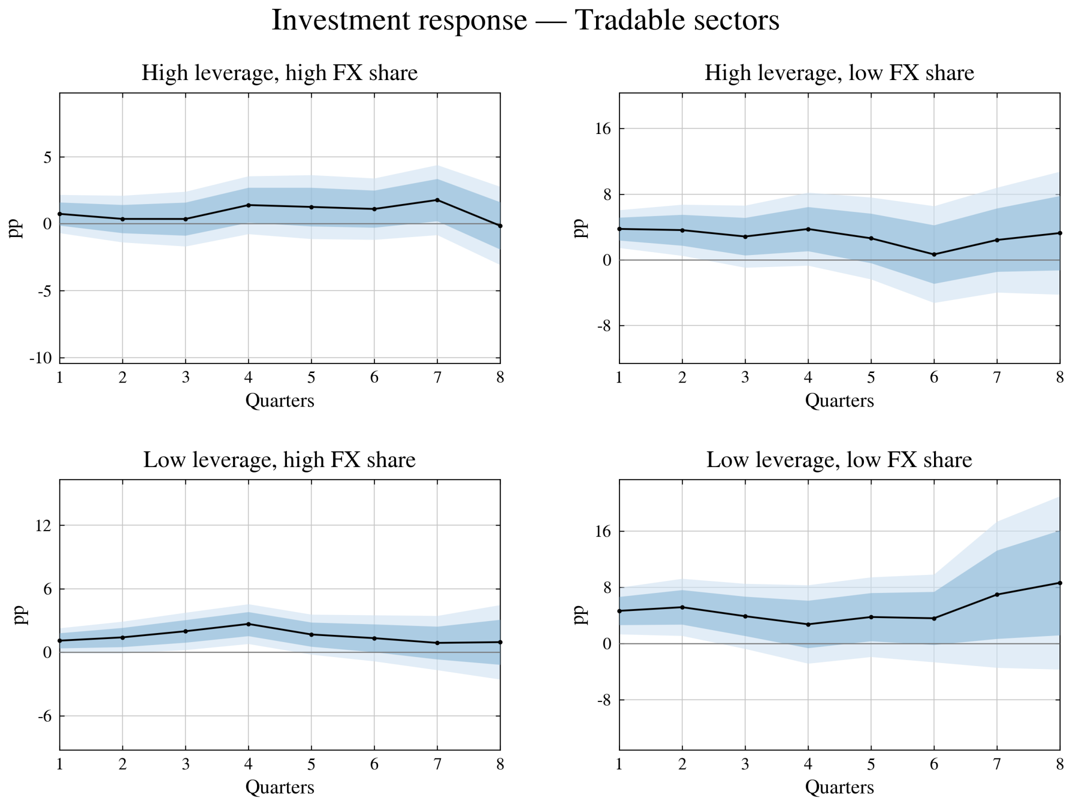
Standard errors in parentheses

Standard errors clustered by country × time.

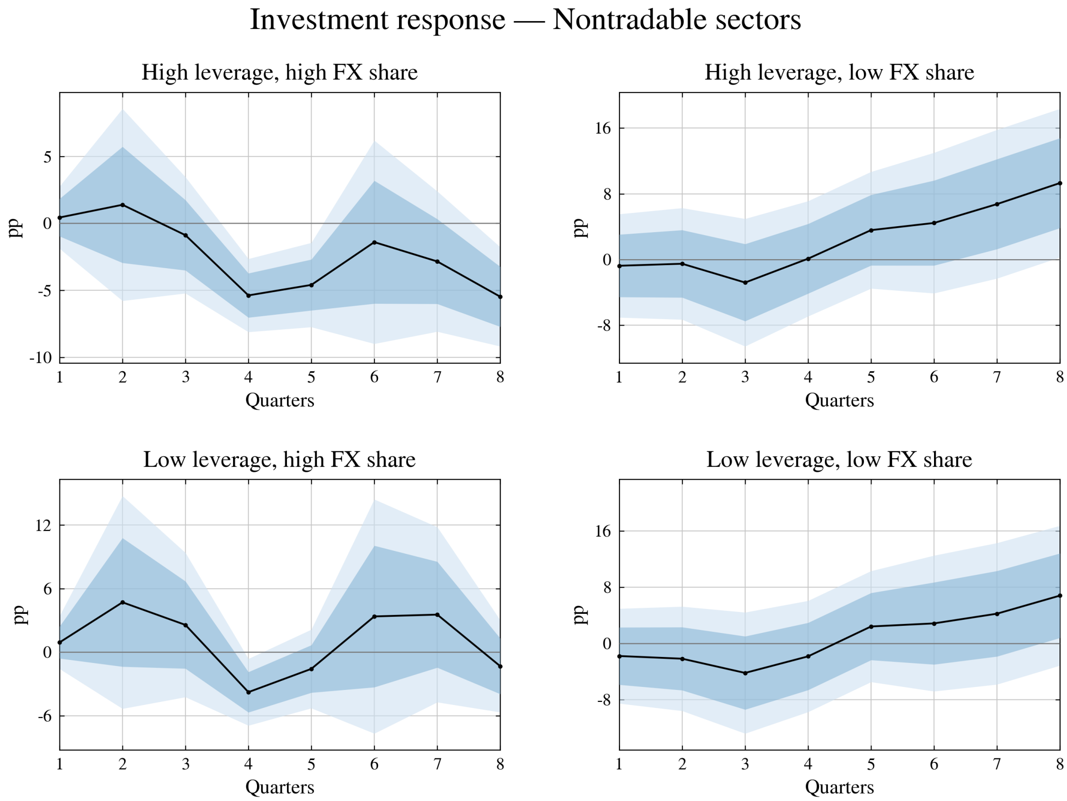
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.4 Tradable and Nontradable Sectors

Figures A1 and A2 plot the marginal effect of a 1pp monetary easing on investment for firms in tradable and nontradable sectors, as discussed in Section 4.2. The two figures are not directly comparable. They evaluate the marginal effect at different exposure values, chosen from the distribution within each group, and the nontradable specification includes firm fixed effects in addition to the country and sector-quarter fixed effects included in both groups. A formal comparison would require common exposure values, the same fixed effects, and a direct test of the difference in responses.

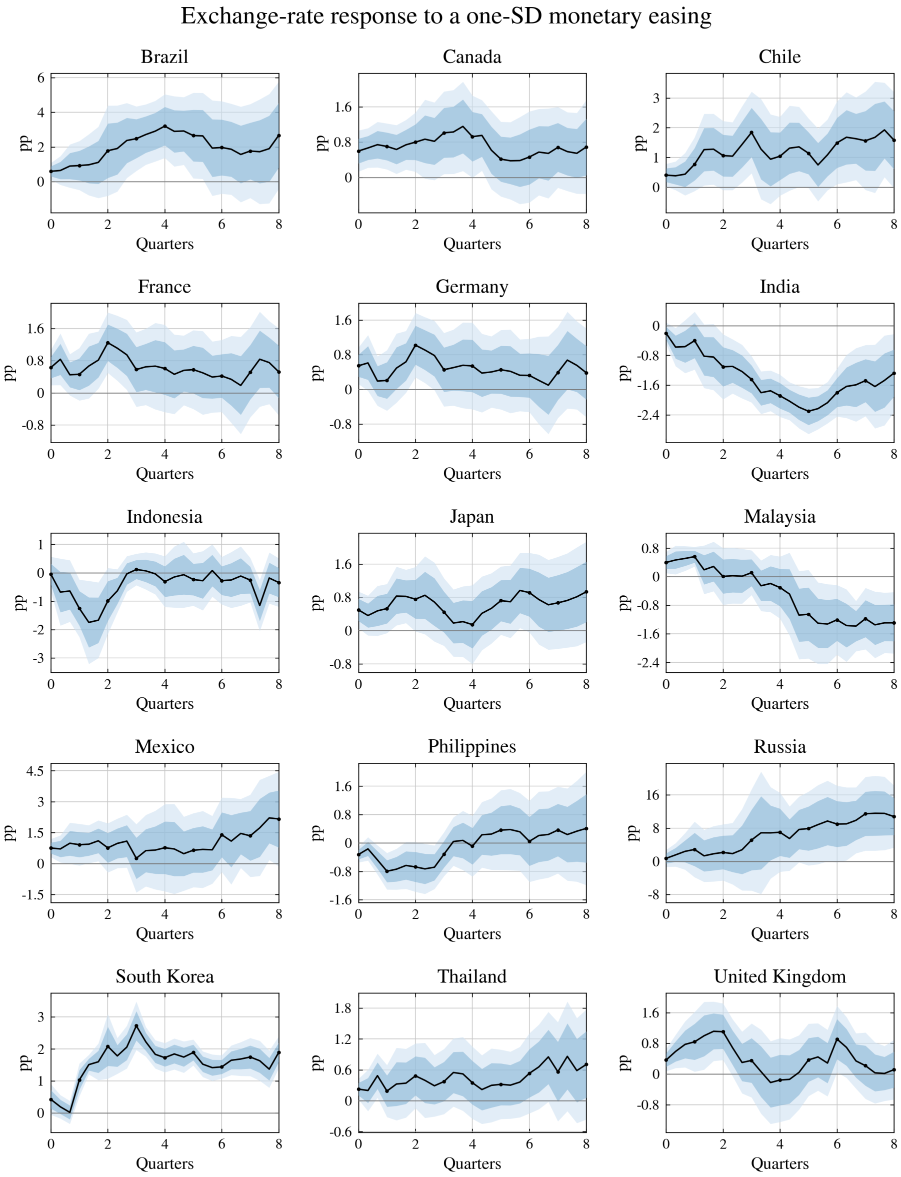


Appendix Figure A1. Investment responses to a 1pp monetary easing in tradable sectors. Low and high FX shares are 0 and 1; low and high standardized leverage are -0.38 and 0.15. Responses are in percentage points. Dark blue areas show approximate 68% confidence bands; light blue areas show 90% confidence bands. The specification includes country and sector-quarter fixed effects.



Appendix Figure A2. Investment responses to a 1pp monetary easing in nontradable sectors. Low and high FX shares are 0 and 0.8; low and high standardized leverage are -0.29 and 0.23. Responses are in percentage points. Dark blue areas show approximate 68% confidence bands; light blue areas show 90% confidence bands. The specification includes firm, country and sector-quarter fixed effects.

A.5 Exchange-Rate Responses to Monetary Easing



Appendix Figure A3. Exchange-rate responses to a country-specific one-standard-deviation monetary easing. Monthly responses are shown over eight quarters, in units of 100 times the cumulative log exchange-rate change. Dark and light blue areas show 68% and 90% confidence bands. Positive values denote depreciation of the local currency.

B Model Appendix

Throughout this appendix, I write the firm's real proceeds from peso and dollar borrowing as

$$L \equiv \frac{b'}{1+i_t}, \qquad F \equiv \frac{e_t\, b^{*\prime}}{1+i^*},$$

so that the issuance cost is $\mathcal{H}_t(b', b^{*\prime}, k') = \mathcal{H}(L, F, k') = \frac{\eta}{2}L^2 + \frac{\eta^*(k')}{2}F^2 + c_F\,\mathbb{1}\{F > 0\}$ and the collateral constraint (11) reads $(1+i_t)L + (1+i^*)F \le \theta_c\, q_t k'$.

B.1 Firms' Recursive Problem

One debt state. In a perfect-foresight path, a firm choosing its debt at t knows Π_{t+1} and e_{t+1}. Its real repayment due at $t+1$ is therefore known at t:

$$R' = \frac{b'}{\Pi_{t+1}} + e_{t+1}b^{*\prime} = \frac{L}{\varrho_t} + \frac{G_{t+1}}{\varrho^*}F, \qquad \varrho_t \equiv \frac{\Pi_{t+1}}{1+i_t}, \quad \varrho^* \equiv \frac{1}{1+i^*}, \quad G_{t+1} \equiv \frac{e_{t+1}}{e_t}, \tag{27}$$

where ϱ_t equals the household's real discount factor $\Lambda_{t,t+1}$ by (17). Cash on hand depends on debt only through R: $x = \pi_t(k,z) + q_t(1-\delta)k - R$. The firm's problem can therefore be written with the state (k, R, z):

$$\begin{aligned} V_t(k,R,z) &= \chi\big[x - \mathcal{C}(x)\big] + (1-\chi)\,W_t(x,k,z), \\ W_t(x,k,z) &= \max_{k',R'} \Big\{ d - \mathcal{C}(d) + \varrho_t\,\mathbb{E}\big[V_{t+1}(k',R',z') \,\big|\, z\big] \Big\}, \\ d &= x - q_t\big[k' + \Psi_k(k',k)\big] + P_t^{\text{net}}(R',k'), \end{aligned} \tag{28}$$

where $P_t^{\text{net}}(R',k')$ is the largest net proceeds that deliver the promised repayment R':

$$\begin{aligned} P_t^{\text{net}}(R',k') = &\max_{L,F\ge 0}\; L + F - \mathcal{H}(L,F,k') \\ &\text{s.t.} \quad \frac{L}{\varrho_t} + \frac{G_{t+1}}{\varrho^*}F = R', \quad (1+i_t)L + (1+i^*)F \le \theta_c\, q_t k'. \end{aligned} \tag{29}$$

The inner problem (29) does not involve the continuation value. Within each branch ($F = 0$ or $F > 0$) its solution is in closed form. When the collateral constraint is slack and both currencies are used, the multiplier ξ on the repayment constraint satisfies

$$1-\eta L=\frac{\xi}{\varrho_t},\qquad 1-\eta^*(k')F=\frac{\xi\,G_{t+1}}{\varrho^*},\qquad \xi=\frac{\frac{1}{\varrho_t\eta}+\frac{G_{t+1}}{\varrho^*\eta^*(k')}-R'}{\frac{1}{\varrho_t^2\eta}+\frac{G_{t+1}^2}{\varrho^{*2}\eta^*(k')}},$$

and $\partial P_t^{\text{net}}/\partial R'=\xi$. When the collateral constraint binds, the firm chooses F along the line $(1+i_t)L+(1+i^*)F=\theta_c\,q_t k'$. The outer choice of R' solves its first-order condition, using the derivative of V_{t+1} with respect to R implied by the envelope condition. Net proceeds also depend on k', through the collateral constraint and through the depth of the dollar market. When the collateral constraint is slack, the envelope theorem gives $\partial P_t^{\text{net}}/\partial k'=\frac{b}{2}\eta^* k'^{-b-1}F^2$, which enters the first-order condition for k'.

Date-0 revaluation. Debt inherited at the date of an unanticipated shock was contracted at expected prices. With $b\equiv B/P_{-1}$, real repayment at $t=0$ is $R_0=b/\Pi_0+e_0 b^*$. Surprise inflation lowers the real value of inherited peso debt, and a surprise real depreciation raises the real value of inherited dollar debt. These are the two revaluation effects in Appendix B.3.

Taste shocks. As in Hegarty et al. (2024), I add independent extreme-value taste shocks to smooth the discrete choices on the numerical grid. With v_n^0 and v_n^1 the values of borrowing only in pesos and of borrowing in dollars at capital choice k_n', and σ and σ_b the scales of the taste shocks, the continuation value is

$$W=\sigma\ln\sum_n\exp\left(\frac{u_n}{\sigma}\right),\qquad u_n=\sigma_b\ln\left[\exp\left(\frac{v_n^0}{\sigma_b}\right)+\exp\left(\frac{v_n^1}{\sigma_b}\right)\right],$$

and the choice probabilities are nested logit. The taste shocks are a smoothing device. They are not resources and do not enter any budget constraint. With taste shocks, policies are probabilities over (k_n',branch). The indicator in the law of motion (21) is then replaced by these choice probabilities. On the numerical grid, the continuous repayment R' is assigned to the two neighbouring grid points with weights that preserve its mean (a lottery).

B.2 The Currency Choice

Let $V_x \equiv \partial V/\partial x = \chi[1+\mathcal{C}'(x)] + (1-\chi)\lambda$ denote the marginal value of cash on hand, where $\lambda = 1+\mathcal{C}'(d)$, and let ξ_c denote the multiplier on the collateral constraint (11). Next-period cash on hand falls by $(1+i_t)/\Pi_{t+1}$ per unit of L and by $(1+i^*)\,e_{t+1}/e_t$ per unit of F. The first-order conditions for a firm that borrows in both currencies are

$$\lambda_t(1-\eta L) = (1+i_t)\,\mathbb{E}_t\left[\frac{\Lambda_{t,t+1}V_{x,t+1}}{\Pi_{t+1}}\right] + (1+i_t)\,\xi_c,$$
$$\lambda_t(1-\eta^*(k')F) = (1+i^*)\,\mathbb{E}_t\left[\Lambda_{t,t+1}V_{x,t+1}\frac{e_{t+1}}{e_t}\right] + (1+i^*)\,\xi_c.$$

With a slack constraint ($\xi_c = 0$) the ratio of the two is

$$\frac{1-\eta L}{1-\eta^*(k')F} = \frac{(1+i_t)\,\mathbb{E}_t\left[\Lambda_{t,t+1}V_{x,t+1}/\Pi_{t+1}\right]}{(1+i^*)\,\mathbb{E}_t\left[\Lambda_{t,t+1}V_{x,t+1}\,e_{t+1}/e_t\right]}.$$

Under aggregate uncertainty this ratio contains the covariance between the firm's marginal value of funds and the returns on each currency, which is the hedging motive behind currency choice in Salomao and Varela (2021). In a perfect-foresight path, the firm-specific expectations cancel, and using $\Pi_{t+1}e_{t+1}/e_t = \mathcal{E}_{t+1}/\mathcal{E}_t$ gives (26).

Selection into foreign currency borrowing. The fixed cost generates a threshold in borrowing needs. Consider the steady state ($i = i^*$, $G = 1$) and a firm raising total proceeds $T = L + F$ with a slack collateral constraint. Borrowing only in pesos costs $\frac{\eta}{2}T^2$. The optimal split, $F = \frac{\eta}{\eta+\eta^*(k')}T$, costs $\frac{\eta\eta^*(k')}{2(\eta+\eta^*(k'))}T^2$. The saving from accessing the dollar market is $\frac{\eta^2}{2(\eta+\eta^*(k'))}T^2$, so the firm pays c_F only if

$$T > \frac{\sqrt{2c_F(\eta+\eta^*(k'))}}{\eta}.$$

Firms with high borrowing needs, namely large firms and productive firms that are growing, select into dollar debt. For $b > 0$, the threshold also falls with the firm's capital, and the dollar share of a dollar borrower rises with it, because larger firms face a deeper dollar market. In the dynamic model the threshold also depends on the firm's marginal value of funds.

B.3 Monetary Easing and Firm Balance Sheets

This subsection characterizes how an unanticipated monetary easing affects firms' investment through their balance sheets. Throughout, Δ denotes the deviation at date t from the path expected at $t-1$, caused by an unanticipated easing at t.

Revaluation of debt due. Debt due at t, $R_{i,t} \equiv b_{i,t}/\Pi_t + e_t b^*_{i,t}$ with $b_{i,t} = B_{i,t}/P_{t-1}$, was contracted at $t-1$, so an easing that raises the price level and depreciates the peso revalues it. Let $\text{Lev}_{i,t} \equiv R_{i,t}/(q_t k_{i,t})$ denote leverage and $\text{FX}_{i,t} \equiv e_t b^*_{i,t}/R_{i,t}$ the dollar share of debt due. Holding the inherited debt $(b_{i,t}, b^*_{i,t})$ fixed, the revaluation changes cash on hand (7) by

$$\frac{\Delta^R x_{i,t}}{q_t k_{i,t}} = -\text{Lev}_{i,t}\Big[\text{FX}_{i,t}\,\Delta \ln e_t - \big(1-\text{FX}_{i,t}\big)\Delta \ln \Pi_t\Big] \tag{30}$$

to a first order. Surprise inflation erodes the real burden of peso debt, and the real depreciation raises the burden of dollar debt. Since $e_t = \mathcal{E}_t P^*_F/P_t$, with P^*_F constant and P_{t-1} predetermined, $\Delta \ln e_t = \Delta \ln \mathcal{E}_t - \Delta \ln \Pi_t$, so

$$\Delta \ln e_t + \Delta \ln \Pi_t = \Delta \ln \mathcal{E}_t. \tag{31}$$

Shifting debt due from pesos to dollars therefore exposes the firm to the *nominal* depreciation: the firm bears the real depreciation on dollar debt and forgoes the inflation erosion of peso debt.

From cash on hand to investment. Let

$$m_{i,t} \equiv q_t \frac{\partial k_{i,t+1}}{\partial x_{i,t}}$$

denote the firm's marginal propensity to invest out of cash on hand: the value of additional capital the firm buys with an additional unit of cash on hand. It is a property of the firm's capital policy, evaluated at its state. The revaluation changes capital by $\Delta^R \ln k_{i,t+1} \approx (\partial k_{i,t+1}/\partial x_{i,t})\, \Delta^R x_{i,t}/k_{i,t+1}$, so by (30)

$$\Delta^R \ln k_{i,t+1} \approx -m_{i,t} \frac{k_{i,t}}{k_{i,t+1}} \mathrm{Lev}_{i,t} \Big[\mathrm{FX}_{i,t} \Delta \ln e_t - \big(1 - \mathrm{FX}_{i,t}\big) \Delta \ln \Pi_t \Big]. \tag{32}$$

The rest of the response, $\Delta \ln k_{i,t+1} - \Delta^R \ln k_{i,t+1}$, reflects the effects of the easing on operating earnings, the price of capital, interest rates and expected depreciation.

The propensity $m_{i,t}$ depends on the firm's financial position. Lemma 1 below shows the following.

- *Dividend payers* ($d_{i,t} > 0$): $m_{i,t} = 0$. The marginal value of internal funds is one, cash on hand does not enter the firm's first-order conditions, and a fall in cash on hand lowers dividends one for one.
- *Equity issuers* ($d_{i,t} < 0$): $m_{i,t} \to 0$ as $\varphi_2 \to 0$. The firm replaces lost cash on hand with equity at a marginal cost $1 + \varphi_1 + \varphi_2 |d_{i,t}|$ that rises only at rate φ_2.
- *Firms at the dividend kink* ($d_{i,t} = 0$, collateral constraint slack): $0 \leq m_{i,t} \leq q_t / \big(q_t(1 + \Psi_{k,1}) + \mathcal{H}_{t,k}\big)$, where $\Psi_{k,1}$ and $\mathcal{H}_{t,k}$ are the derivatives of the adjustment and issuance costs with respect to $k_{i,t+1}$. Because equity costs φ_1 at the margin, these firms neither pay dividends nor issue equity, and they split a fall in cash on hand between less investment and more borrowing.
- *Collateral-constrained firms* ($d_{i,t} = 0$, collateral constraint binds): $m_{i,t} = q_t / \big(q_t[1 - \theta_c/(1 + \tilde{\imath}_{i,t})] + q_t \Psi_{k,1} + \mathcal{H}'_c\big)$, where $1/(1 + \tilde{\imath}_{i,t})$ is the firm's average bond price and $\mathcal{H}'_c$ the marginal issuance cost along the constraint. Up to the marginal adjustment and issuance costs, $m_{i,t} \approx 1/\big(1 - \theta_c/(1 + \tilde{\imath}_{i,t})\big) > 1$: each unit of cash on hand supports a multiple of capital, because debt finances a fraction $\theta_c/(1 + \tilde{\imath}_{i,t})$ of each unit.

The balance-sheet effect therefore operates only through firms whose marginal value of internal funds exceeds one, and it is strongest for firms at the collateral constraint.

The interaction of leverage and foreign currency debt.

Proposition 1. *Consider an unanticipated easing that depreciates the peso,* $\Delta \ln \mathcal{E}_t > 0$. *Holding* $m_{i,t}$ *and* $k_{i,t+1}/k_{i,t}$ *fixed,*

$$\frac{\partial\, \Delta^R \ln k_{i,t+1}}{\partial\, \mathrm{FX}_{i,t}} \approx -\, m_{i,t}\, \frac{k_{i,t}}{k_{i,t+1}}\, \mathrm{Lev}_{i,t}\, \Delta \ln \mathcal{E}_t \leq 0,$$
$$\frac{\partial^2\, \Delta^R \ln k_{i,t+1}}{\partial\, \mathrm{FX}_{i,t}\, \partial\, \mathrm{Lev}_{i,t}} \approx -\, m_{i,t}\, \frac{k_{i,t}}{k_{i,t+1}}\, \Delta \ln \mathcal{E}_t \leq 0,$$

with strict inequalities whenever $m_{i,t} > 0$.

Proof. Differentiate (32) with respect to $\mathrm{FX}_{i,t}$ and use (31). Then differentiate the result with respect to $\mathrm{Lev}_{i,t}$. $\square$

For collateral-constrained firms, the balance-sheet effect is also convex in leverage, because a highly levered firm has little internal funds relative to its capital and therefore a small planned capital stock relative to the loss.

Corollary 1. *For a collateral-constrained firm, neglecting the marginal adjustment and issuance costs and holding the dollar share of new debt fixed,*

$$\Delta^R \ln k_{i,t+1} \approx -\frac{\mathrm{Lev}_{i,t}}{a_{i,t} - \mathrm{Lev}_{i,t}} \Big[\mathrm{FX}_{i,t}\, \Delta \ln e_t - \big(1 - \mathrm{FX}_{i,t}\big) \Delta \ln \Pi_t \Big],$$

where $a_{i,t} \equiv \pi_{i,t}/(q_t k_{i,t}) + (1-\delta) - (q_t \Psi_k + \mathcal{H}_t)/(q_t k_{i,t})$. *Holding* $a_{i,t}$ *fixed,*

$$\frac{\partial\, \Delta^R \ln k_{i,t+1}}{\partial\, \mathrm{FX}_{i,t}} \approx -\frac{\mathrm{Lev}_{i,t}}{a_{i,t} - \mathrm{Lev}_{i,t}}\, \Delta \ln \mathcal{E}_t,$$
$$\frac{\partial^2\, \Delta^R \ln k_{i,t+1}}{\partial\, \mathrm{FX}_{i,t}\, \partial\, \mathrm{Lev}_{i,t}} \approx -\frac{a_{i,t}}{(a_{i,t} - \mathrm{Lev}_{i,t})^2}\, \Delta \ln \mathcal{E}_t.$$

Proposition 1 has three implications. First, a higher dollar share lowers the investment response to an easing, in proportion to the nominal depreciation the easing causes. Second, this penalty rises with leverage, which is the negative triple interaction β_6 in Section 4. Corollary 1 shows that for constrained firms the penalty rises more than proportionally with leverage. Third, the penalty arises only for firms with $m_{i,t} > 0$, that is, firms at the dividend kink or at the collateral constraint. Because more levered firms are more likely to be in these positions, the cross-section of financial positions reinforces the interaction. For firms that borrow only in pesos, (32) implies that surprise inflation raises the investment response in proportion to leverage, $m_{i,t}(k_{i,t}/k_{i,t+1})\text{Lev}_{i,t}\Delta \ln \Pi_t$, which contributes to the positive coefficient on leverage, β_5.

Offsetting forces. Two features of the model limit the balance-sheet effect. First, debt is one-period, so the easing revalues only the debt due at the date of the shock, while lower interest rates persist. Second, after the initial depreciation the peso is expected to appreciate. The peso cost of a dollar borrowed at t is $(1+i^*)\mathcal{E}_{t+1}/\mathcal{E}_t$, so the expected appreciation lowers the cost of new dollar debt, and firms that borrow in dollars refinance more cheaply after the initial loss. These forces explain why, in the quantitative model, foreign currency debt weakens the investment response of highly levered firms rather than reversing it.

Equation (30). Cash on hand is $x_{i,t} = \pi_{i,t} + q_t(1-\delta)k_{i,t} - R_{i,t}$. Holding $(b_{i,t}, b^*_{i,t})$ fixed, a first-order change in (Π_t, e_t) changes debt due by

$$\Delta R_{i,t} \approx -\frac{b_{i,t}}{\Pi_t}\,\Delta \ln \Pi_t + e_t b^*_{i,t}\,\Delta \ln e_t = R_{i,t}\Big[\text{FX}_{i,t}\,\Delta \ln e_t - \big(1-\text{FX}_{i,t}\big)\Delta \ln \Pi_t\Big],$$

using $(b_{i,t}/\Pi_t)/R_{i,t} = 1 - \text{FX}_{i,t}$. The revaluation component of cash on hand is $\Delta^R x_{i,t} = -\Delta R_{i,t}$. Dividing by $q_t k_{i,t}$ gives (30).

Notation for the lemma. As at the start of this appendix, write the proceeds of new debt as $L \equiv b_{i,t+1}/(1+i_t)$ and $F \equiv e_t b^*_{i,t+1}/(1+i^*)$, so that $\mathcal{H}_t = \frac{\eta}{2}L^2 + \frac{\eta^*(k')}{2}F^2 + c_F \mathbb{1}\{F > 0\}$ with $k' \equiv k_{i,t+1}$. The cash-flow constraint (8) is

$$q_t\big[k' + \Psi_k(k',k)\big] + \mathcal{H}_t + d = x + L + F. \tag{33}$$

Its partial derivatives are $\mathcal{H}_{t,L} = \eta L$, $\mathcal{H}_{t,F} = \eta^*(k')F$ and $\mathcal{H}_{t,k} = -\frac{b}{2}\eta^* k'^{-b-1}F^2 = -\frac{b}{2}\eta^*(k')F^2/k'$. The continuation value does not depend on x, and the firm's objective is $d - \mathcal{C}(d) + \mathbb{E}_t[\Lambda_{t,t+1}V_{t+1}]$. The derivative of $d - \mathcal{C}(d)$ with respect to d is 1 for $d > 0$ and $1 + \varphi_1 + \varphi_2|d|$ for $d < 0$. At $d = 0$ the function has a kink, with left derivative $1 + \varphi_1$ and right derivative 1.

Lemma 1. *Consider a local change in x that keeps the firm in the same financial position and the same currency branch ($F > 0$ or $F = 0$).*

1. *If $d > 0$, then $m = 0$.*
2. *If $d < 0$ and $\varphi_2 = 0$, then $m = 0$.*
3. *If $d = 0$, the collateral constraint is slack, and capital and the proceeds of each type of debt are normal in cash on hand ($\partial k'/\partial x \geq 0$, $\partial L/\partial x \leq 0$, $\partial F/\partial x \leq 0$), then*

   $$0 \leq m \leq \frac{q_t}{q_t(1 + \Psi_{k,1}) + \mathcal{H}_{t,k}}.$$

   If in addition $k' \geq (1-\delta)k$, so that $\Psi_{k,1} \geq 0$, and $b\,\theta_c < 2$, the upper bound is at most $1/(1 - b\theta_c/2)$.
4. *If $d = 0$ and the collateral constraint binds, then, holding the dollar share of new debt fixed,*

   $$m = \frac{q_t}{q_t\big[1 - \theta_c/(1 + \tilde{\imath}_{i,t})\big] + q_t\Psi_{k,1} + \mathcal{H}'_c},$$

   where $\mathcal{H}'_c$ is the derivative of $\mathcal{H}_t$ with respect to k' along the binding constraint.

Proof. Parts 1 and 2. If $d > 0$, substituting (33) into the objective gives $x + \big[L + F - q_t(k' + \Psi_k) - \mathcal{H}_t\big] + \mathbb{E}_t[\Lambda_{t,t+1}V_{t+1}]$. Cash on hand enters additively and does not appear in the collateral constraint, so the optimal $(k', b', b^{*\prime})$ do not depend on x, and $m = 0$. If $d < 0$ and $\varphi_2 = 0$, then $d - \mathcal{C}(d) = (1 + \varphi_1)d$, the objective is $(1 + \varphi_1)\big(x + [\cdots]\big) + \mathbb{E}_t[\Lambda_{t,t+1}V_{t+1}]$, and the same argument applies.

Part 3. With $d = 0$, differentiating (33) with respect to x gives

$$\left[q_t(1+\Psi_{k,1}) + \mathcal{H}_{t,k}\right]\frac{\partial k'}{\partial x} = 1 + (1-\eta L)\frac{\partial L}{\partial x} + \big(1-\eta^*(k')F\big)\frac{\partial F}{\partial x}.$$

At an optimum, the marginal net proceeds of each type of debt the firm issues are positive, $1-\eta L > 0$ and $1-\eta^*(k')F > 0$, by the first-order conditions in Appendix B.2, whose right-hand sides are positive. Under normality, the last two terms are nonpositive, so $\partial k'/\partial x \leq 1/\big(q_t(1+\Psi_{k,1}) + \mathcal{H}_{t,k}\big)$ whenever the denominator is positive, and $\partial k'/\partial x \geq 0$ by assumption. For the bound on the denominator, $\eta^*(k')F < 1$ implies $|\mathcal{H}_{t,k}| = \frac{b}{2}\eta^*(k')F^2/k' < \frac{b}{2}F/k'$. Since $b_{i,t+1} \geq 0$, the collateral constraint implies $(1+i^*)F = e_t b^*_{i,t+1} \leq \theta_c q_t k'$, so $F/k' \leq \theta_c q_t$. Hence $|\mathcal{H}_{t,k}| < \frac{b}{2}\theta_c q_t$, and with $\Psi_{k,1} \geq 0$ the denominator exceeds $q_t(1-b\theta_c/2) > 0$. Multiplying by q_t gives the bounds on m.

Part 4. With a binding constraint, $b_{i,t+1} + e_t b^*_{i,t+1} = \theta_c q_t k'$. Let s' denote the dollar share of new debt. Then $L + F = (b_{i,t+1} + e_t b^*_{i,t+1})\big[(1-s')/(1+i_t) + s'/(1+i^*)\big] = \theta_c q_t k'/(1+\tilde{\imath}_{i,t})$. Substituting into (33) with $d = 0$ gives

$$q_t\left[1 - \frac{\theta_c}{1+\tilde{\imath}_{i,t}}\right]k' + q_t\Psi_k(k',k) + \mathcal{H}_t = x.$$

Holding s' fixed, L and F are proportional to k', so $\mathcal{H}_t$ depends on x only through k'. Differentiating with respect to x gives the expression for m. □

Corollary 1. Neglecting $q_t\Psi_{k,1}$ and $\mathcal{H}'_c$, Part 4 gives $m = 1/\big(1-\theta_c/(1+\tilde{\imath}_{i,t})\big)$, and the displayed equation in its proof gives $q_t k' = m\,\tilde{x}_{i,t}$ with $\tilde{x}_{i,t} \equiv x_{i,t} - q_t\Psi_k - \mathcal{H}_t$. Hence $m\,k_{i,t}/k_{i,t+1} = q_t k_{i,t}/\tilde{x}_{i,t}$. Using the definition of cash on hand, $\tilde{x}_{i,t}/(q_t k_{i,t}) = a_{i,t} - \mathrm{Lev}_{i,t}$, where $a_{i,t}$ does not depend on debt due. Substituting into (32) gives the first expression. Differentiating with respect to $\mathrm{FX}_{i,t}$ and using (31) gives the second. Finally, $\partial\big[\mathrm{Lev}/(a-\mathrm{Lev})\big]/\partial\mathrm{Lev} = a/(a-\mathrm{Lev})^2$ gives the third.

Remark. *These are first-order, partial-equilibrium statements. They hold the firm's financial position fixed and treat $m_{i,t}$ and $k_{i,t+1}/k_{i,t}$ as given. In the model, both vary with leverage and with the firm's position relative to the dividend kink and the collateral constraint, and general* *equilibrium prices respond to the easing. Section 6 accounts for both.*

B.4 Retailers

Retailer j chooses $\{P_{j,t}\}$ to maximize $\mathbb{E}_0 \sum_t \Lambda_{0,t}\, p_{H,t}\big[(P_{j,t}/P_{H,t})^{1-\varepsilon_p} - m_t(P_{j,t}/P_{H,t})^{-\varepsilon_p} - \frac{\psi_p}{2}(P_{j,t}/P_{j,t-1} - 1)^2\big]Y_t$, given demand $y_{j,t} = (P_{j,t}/P_{H,t})^{-\varepsilon_p}Y_t$. Profits are measured in units of the consumption bundle. The first-order condition evaluated at the symmetric equilibrium $P_{j,t} = P_{H,t}$ gives (14). Retailer profits, $p_{H,t}Y_t\big[1 - m_t - \frac{\psi_p}{2}(\Pi_{H,t} - 1)^2\big]$, are rebated to households.

B.5 Walras' Law

This subsection shows that the household budget constraint (16) is implied by the other equilibrium conditions. Divide (16) by P_t. Net cash flows to households are

$$\begin{aligned} D_t = {} & \underbrace{(1-\chi)\int \big[d_t - \mathcal{C}(d_t)\big]\,d\mu_t + \chi\int \big[x_t - \mathcal{C}(x_t)\big]\,d\mu_t - q_t\chi k_0}_{\text{firms}} \\ & + \underbrace{p_{H,t}Y_t\Big[1 - m_t - \tfrac{\psi_p}{2}(\Pi_{H,t} - 1)^2\Big]}_{\text{retailers}} + \underbrace{q_t I_t - p_{H,t}\Phi(I_t, K_t)}_{\text{capital producer}}. \end{aligned}$$

Substitute the firm cash-on-hand and cash-flow definitions (7) and (8). Use $\int \pi_t\, d\mu_t = (1-\nu)m_t p_{H,t}Y_t$ and $w_t N_t = \nu m_t p_{H,t} Y_t$. Use the capital-goods identity (c) and peso-bond clearing (d). The household budget then becomes

$$C_t + \mathcal{Q}_t + p_{H,t}\Phi(I_t, K_t) + p_{H,t}\tfrac{\psi_p}{2}(\Pi_{H,t} - 1)^2 Y_t = p_{H,t}Y_t - e_t\int b^*\, d\mu_t + (1-\chi)\int \frac{e_t\, b_t^{*\prime}}{1+i^*}\, d\mu_t.$$

This is the economy-wide budget constraint. Multiply (24) by $p_{H,t}$, use $p_{H,t}\cdot \varpi p_{H,t}^{-\varepsilon} + e_t\cdot(1-\varpi)e_t^{-\varepsilon} = 1$ from the CPI identity, and subtract the balance of payments (25). The result is the same expression. Hence, given (a)–(d), the home-good market and the balance of payments imply the household budget constraint, and vice versa.

B.6 Solution Method

I solve for the stationary equilibrium and compute the economy's response to monetary policy shocks using the sequence-space Jacobian method of Auclert et al. (2021a).

- **Firm block.** The firm block is solved on a grid over (k, R, z). The capital grid coincides with the capital choice grid. The repayment grid is defined relative to capital, R/k, so that firms of every size have the same resolution in leverage.
- **Backward step.** Given (k', branch), the repayment R' solves its first-order condition to machine precision. V_{t+1} is interpolated with cubic Hermite polynomials using the envelope derivative V_R. This makes policies smooth in prices, which the numerical differentiation of the backward step requires.
- **Firm-block Jacobians.** These are computed with the fake-news algorithm, with respect to Θ_t, ϱ_t, G_{t+1}, q_t, the nominal bond price $1/(1+i_t)$ that enters the collateral constraint, and the two date-0 revaluations. I verify them column by column against direct perturbations of the nonlinear transition.
- **Large shocks.** Nonlinear perfect-foresight transitions are computed by Newton iteration on the general equilibrium residuals, using the linearized Jacobian.